

UNIVERSITÉ DE LORRAINE

MASTER THESIS

Continual Learning for Hate Speech Detection

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Declaration of Authorship

I, Alberto LORENTE GALÉ, declare that this thesis titled, “Continual Learning for Hate Speech Detection” and the work presented in it are my own. I confirm that:

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- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
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Abstract

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MSc. in Natural Language Processing

Continual Learning for Hate Speech Detection

by Alberto LORENTE GALÉ

Hate speech, defined as "the communications of animosity or disparagement of an individual or a group on account of a group characteristic such as race, color, national origin, sex, disability, religion, or sexual orientation" (Nockleby, 2000) in the Encyclopedia of the American Constitution, has been a popular topic within Natural Language Processing (NLP) since the beginning of the social media era. As language changes and new hate speech domains appear due to sociopolitical changes, there is a need for models that can adapt quickly and effectively. Continual Learning, the subfield of Machine Learning that focuses on learning tasks incrementally, remains underexplored despite its adequacy for the task. In this paper I explore traditional continual learning methods, EWC, Lwf, MAS and A-GEM, for language models for Hate Speech Detection and conclude that A-GEM can effectively tackle datastreams and achieve better performance detecting hateful messages for a variety of experiments.

The code is available at [Continual Hate](#).

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Chapter 1

Introduction

The present Master's Thesis was carried out in the context of the **DesCartes program** hosted by **CNRS@CREATE** under the supervision of **Farah Benamara**, Co-lead of WP5: *Hybrid AI for Linguistically-Informed Processing of Texts & Conversations*.

The DesCartes program was established in 2021 as a partnership between France and Singapore with Nanyang Technological University (NTU), National University of Singapore (NUS), A*STAR and CNRS among the main collaborators. Its research focus is **Intelligent Modelling for Decision-making in Critical Urban Systems** and smart cities with an emphasis on digital twins.

Regardless of recent de-regulation trends, the need for robust content moderation tools is evident in the context of citizens interacting with each other or official governing channels through different social media platforms that act as digital town squares.

Hate speech, defined as "the communications of animosity or disparagement of an individual or a group on account of a group characteristic such as race, color, national origin, sex, disability, religion, or sexual orientation" Nockleby, 2000 in the *Encyclopedia of the American Constitution*, has been a popular topic within Natural Language Processing (NLP) due to the boom of user-generated web content Schmidt and Wiegand, 2017.

Although the vast majority of research in the field focuses on Hate Speech Detection (HSD) as a standard fine-tuning binary or multi-label classification task (as in Wijanarko et al., 2024) or, more recently, as an In-Context Learning (ICL) prompting task (as in Ghorbanpour et al., 2025 and Roy et al., 2023), the topic lends itself quite seamlessly to less explored learning paradigms such as Active Learning, Curriculum Learning or Continual Learning (CL), the focus of this thesis. The latter's main objective is for models to learn data streams in a sequential way in order to quickly adjust to changing data distributions, incorporate additional classes or learn new tasks altogether while not losing the capabilities they gained in previous training stages.

Why Continual Learning in particular? From a sociolinguistic perspective, any language is bound to go through diachronic evolution and lexical changes which would affect the actual distribution of its latent representation, rendering temporal domain adaptation essential. From a sociopolitical perspective, new emerging types and targets of hate speech may arise due to political circumstances, historical events and demographic shifts. From the perspective of a social media platform, there may be a need to quickly adapt to changing user preferences or regulations with respect to flagged content. Finally, from a practical perspective, one may not have all the data they would like to use to train a model at once or they may not be able to store it to retrain the model from scratch after more data has been collected, forcing them to split the training process in various stages.



FIGURE 1.1: Example of Emerging Hateful Phenomena

Unfortunately, papers concerning Continual Learning and Hate Speech Detection appear sparsely and, when they do, they rarely report their findings with consolidated Continual Learning metrics, opting overwhelmingly to report the performance of their models only after the training stream has been completed. It is also unfortunate that the data stream orders are not usually discussed and their training set ups are rather optimistic. Although not ideal, this phenomena could be explained by the fact that Continual Learning as a discipline was born within the Computer Vision (CV) realm and its established practices have yet to make the jump into the Hate Speech Detection world.

In a previous work, Bourgeade et al., 2023 studied the potential domain transferability among established, widely used datasets, each representing different domains of hate speech, and assessed common challenges and potential issues to consider when doing so. Given that the data collection processes of these kind of datasets can often be fragmented as hate changes over time and task transferability lies at the center of Continual Learning, one of the main goals of this thesis is to determine whether Continual Learning may excel in a situation where a model is being trained in different hate speech domains in an incremental way, mimicking those data collection processes.

All in all, in this thesis I am aiming to:

- Promote the adoption of Continual Learning culture in Hate Speech Detection.
- Implement traditional Continual Learning techniques and study their impact in Hate Speech Detection.
- Assess the feasibility of Continual Learning for Hate Speech Domain Transfer.
- Determine which domains successfully transfer into each other.
- Determine if Continual Learning significantly improves zero-shot performance on unseen datasets.
- Determine if Continual Learning has a positive effect in the detection of hateful messages, specifically.

Chapter 2

Literature Review

2.1 Recent Trends in Hate Speech Detection

Hate speech continues to be a mainstream topic in the social media era. As some social media platforms take a turn towards de-regularizing content moderation, studies continue to warn about the rise of hate speech online (Hickey et al., 2025) and its many potential harms, highlighting the need for effective computational tools.

Recent research on Hate Speech Detection has flourished in various directions. Regarding architectural advancements for PLMs¹, Zhu et al., 2024 proposed a Deep Prompt Multi-task Network (DPMN) for abusive language detection based on a continuous prompt encoder which generates a learnable embedding that is concatenated to the input embedding before being processed by a PLM and, later, the task-specific heads. Jiang, 2025 proposed a Contrastive Learning approach with causality guided hard positive sampling for implicit hate speech detection.

As far as non-standard English Hate Speech Detection, Pokrywka and Jassem, 2024 evaluated transformer models for caste and migration hate speech in Tamil, code-mixing between Tamil-English and English using different scripts and concluded that traditional PLMs such as BERT (Devlin et al., 2019) and RoBERTa (Y. Liu et al., 2019) deeply struggled with Tamil’s script. Wijanarko et al., 2024 set out to monitor hate speech against minorities in Indonesia during the 2024 presidential election through a collaboration with the General Election Supervisory Body in Indonesia (BAWASLU). Finally, Achamaleh et al., 2024 faced the challenge of code-mixed Telugu and English using classic Recurrent Neural Networks (RRNs).

As far as LLMs, there have been a plethora of papers exploring the viability of prompting for Hate Speech Detection (F. Huang et al., 2023), studying how prompt design affects LLMs’ capabilities both in English (Roy et al., 2023) and other languages (Ghorbanpour et al., 2025; Plaza-del-Arco et al., 2023) and the role of class label verbalizers, that is, the way a class label is expressed within a prompt (Plaza-del-Arco et al., 2023). More recently, Yamagishi, 2024 studied zero-shot prompting with LLaVA-1.5 13B (H. Liu et al., 2024) to detect hateful content in the context of the Russia-Ukraine conflict and concluded that more complex prompts that include contextual information and class label definitions did not help the model improve its zero-shot performance.

There has also been a surge of research concerning implicit hate speech and explainability. F. Huang et al., 2023 proposed a Chain of Explanations (CoE) prompting strategy which consisted of inputting text, its label and the group targeted by the text to generate explanations for implicit hate speech tweets. Nirmal et al., 2024 advocated for more interpretable hate speech detection models by using an auxiliary

¹Throughout this thesis, "PLMs" will be used to refer to traditional Transformer-based models and "LLMs" will be used to refer to those language models, usually following decoder-only architectures, with around a billion parameters in size and which exhibit emergent capabilities (Wei et al., 2022)

LLM to produce explanations (in the form of textual features) to use alongside the original text embedding to train a detection model, effectively incorporating interpretability directly into the training process, and consequently the model. Lastly, Yang et al., 2023 proposed a framework to generate Chain of Thought (CoT) (Wei et al., 2023) explanations with LLMs to train Hate Speech Detection classifiers which outperformed models trained solely on human annotations of the implied statements of messages.

Moving to the topic of the impact of alignment in LLMs' sensitivity towards hate speech and abusive language, Zhang et al., 2025 showed that instruction-tuned models like Flan-T5 (Chung et al., 2022) under-predict abusive labels while models that are fine-tuned using Reinforcement Learning with Human Feedback (RLHF), like LLaMA 2-Chat (Touvron, Martin, et al., 2023), become too sensitive and over-predict abusive labels. Jin et al., 2024 found LLMs to exhibit extreme behavior in the classification of implicit hate speech text while exhibiting poor confidence estimation of their predictions assigning similar ranges of confidence for simple and complex datasets.

Although not solely related to Hate Speech Detection, one of the most notable advances in the last few years in terms of content safety is the release of the series of Llama Guard models (Inan et al., 2023; Grattafiori et al., 2024; Chi et al., 2024), fine-tuned Llama models for safety classification of LLM outputs. Their output follows the AI Safety Benchmark taxonomy (Vidgen et al., 2024) which includes a *Hate* category defined as "responses that demean or dehumanize people on the basis of their sensitive, personal characteristics. People who are targeted because of personal attributes that are 'protected' or 'sensitive' characteristics".

Overall, Hate Speech Detection seems to be moving in the direction of implicit hate speech detection, distillation of LLMs' knowledge into simpler and more explainable models, advancements in Hate Speech Detection for low-resourced languages, In-Context Learning studies and aligning the sensitivity of LLMs' to be closer to that of humans.

2.2 Continual Learning and Natural Language Processing

Continual Learning aims to incrementally learn tasks across time, each with their own distinct dataset, in order to adapt to changing environments. Although Continual Learning was born in the Computer Vision space, its techniques have been adopted in certain Natural Language Processing tasks. In one of the first Continual Learning papers proposed for NLP tasks, Sun et al., 2019 introduced LAnguage MOdeling for Lifelong language learning (LAMOL), which relied on generating pseudo-samples of previous Question-Answering task data to replay during later tasks without the need to train a separate generator model. In terms of architecture advances, Ke et al., 2021 proposed a Capsule Network (Sabour et al., 2017) model inspired by Adapter-BERT (Houlsby et al., 2019) for aspect sentiment classification tasks which positively transferred knowledge to both forward and backward tasks and Ke et al., 2022 used parallel adapters and task masks to protect important neurons to train a RoBERTa (Y. Liu et al., 2019) model for continual post-training with unlabeled data, which was afterwards fine-tuned in a non-CL scenario for classification tasks.

On the LLM side, Continual Learning has been proposed for each of the three standard training stages: pre-training, instruction tuning and aligning. Despite the usual sequential workflow that is followed when training LLMs in these three stages,

it is crucial that they can be trained iteratively so that they do not forget how to follow instructions after going through Continual Fine-Tuning, for example (Wu et al., 2024).

Relying on data replay techniques, W. Huang et al., 2025 highlights the challenge of performance rehearsal, data replay, in LLMs whose training data is not publicly available. They address this challenge with Self-Synthesized Rehearsal (SSR). First, the base LLM is used to generate synthetic data via In-Context Learning and the latest LLM is used to refine and correct the outputs of the synthetic data previously generated. In the end, a sample of synthetic data is selected with a clustering algorithm.

Scialom et al., 2022 developed a Continual T0-3B (Sanh et al., 2022) model using memory buffers that stored around 1% of the data. It is remarkable that their Continual-T0 managed to keep its performance in the T0 zero-shot benchmarks even though there was no rehearsal done for the original data.

Other CL techniques relied on modifying Mixture-of-Experts (MoE) (Shazeer et al., 2017) architectures as well as Parameter Efficient Finetuning (PEFT) techniques like LoRA (Hu et al., 2021) or QLoRA (Dettmers et al., 2023). Chen et al., 2023 introduced Lifelong-MoE, a Mixture-of-Experts model based on GLaM (Du et al., 2022) which adds expert layers and regularizations progressively.

On the PEFT side, Song et al., 2023 put forward two Continual Learning Parameter Efficient Tuning (PET) for LLMs, Static ConPET and Dynamic ConPET. The former uses a PEFT module, like LoRA layers, and dynamic replay of samples gathered from all the historical data. The latter uses a MoE-like architecture where a set of task-specific PET modules, a PET module selector and a cache are implemented with a base LLM, LLaMA-7B (Touvron, Lavril, et al., 2023).

2.2.1 Continual Hate Speech Detection

Although Continual Learning is not a popular topic within Hate Speech Detection, there are a few works that have set out to explore the field. Qian et al., 2021 proposed a novel architecture relying on Variational Representation Learning (VRL) and a Load-Balancing Self-Organizing Incremental Neural Network (LB-SOINN) memory module for incremental fine-grained hate speech classification of hate speech groups based on tweets' texts. Both Omrani et al., 2024 and Lin et al., 2024 aimed to establish continual learning frameworks and benchmarks for problematic content detection and debiasing, respectively, and, lastly, Kang et al., 2025 explored Continual Learning with BERT to detect perturbed texts through feature replay of previous perturbation domains to maximize the similarity between the past memory features and the present ones.

Where these works fall short is in adopting some of the standard practices of Continual Learning. In terms of task order, for example, Qian et al., 2021 only considers a second order post hoc to understand better the poor of performance in their first task and their work does not consider streams of different lengths nor report how the performance of a task evolves *before* training on that specific dataset. Omrani et al., 2024 only considered their upstream tasks chronologically but Lin et al., 2024 does interpolate the order of their data streams although only considering sequences of length four.

Moreover, none of the cited works employ any standard Continual Learning metric, such as Average Incremental Average (AIA), Backward Transfer (BWT) or Forward Transfer (FWT). Qian et al., 2021 does report the average performance at intermediate time steps of the training and shows the performance evolution for one

of their models but, for the most part, the works cited opt to assess the final performance of the models after completing the training on the data stream paying little to no attention to the evolution of the performance on the individual tasks themselves.

When it comes to the types of models trained, current works in Continual Hate Speech Detection have only considered PLMs, more specifically, general purpose BERT variants, turning an eye blind to hate-specific models, such as hateBERT (Caselli et al., 2021) or fBERT (Sarkar et al., 2021), and LLMs even though frameworks like the downstream-upstream task set up of Omrani et al., 2024 closely resemble the Continual Pre-training plus Continual Domain Adaptation workflow.

Chapter 3

Hate Speech Domain Transfer via Continual Learning

Continuing the line of work established in Bourgeade et al., 2023, our main goals were to determine whether applying continual learning techniques during streams of datasets concerning different hate speech domains (for instance xenophobia, misogyny or racism) could help Transformer-based models generalize better, improve their zero-shot performance on unseen datasets and better the predictions of the hate class.

To properly frame the performance of the resulting continual Transformers-based models, their performance across datasets is compared to that of smaller LLMs in the 1-4B parameters range in a few and zero-shot prompting setting.

3.1 Models

The PLMs evaluated for our experiments are RoBERTa (Y. Liu et al., 2019), an optimized version of BERT (Devlin et al., 2019), and two BERT variants retrained to identify hate speech and abusive language, fBERT (Sarkar et al., 2021) and Hate-BERT (Caselli et al., 2021).

In order to frame the performance of the selected PLMs, they are compared with LLMs of varying sizes from the Llama3 herd (Grattafiori et al., 2024), Llama 3.1-3B and Llama 3.2-1B, as well as Llama Guard 3-1B, an instruction-tuned Llama-3.2-1B model for content safety classification (Inan et al., 2023).

Model	Output	Domain	Parameter Size	Model Architecture
roberta-base	Structured	General	125M	PLM
fBERT	Structured	English Offensive Language Detection	110M	PLM
Hate-BERT	Structured	English Abusive Language Detection	110M	PLM
Llama 3.2	Unstructured	General	1B	LLM
Llama Guard 3	Structured: safe/unsafe + Vidgen et al., 2024 Taxonomy	Content Safety Classification	1B	LLM
Llama 3.2	Unstructured	General	3B	LLM
Qwen 3	Unstructured	General	1.7B	LLM
Qwen 3	Unstructured	General	4B	LLM

TABLE 3.1: Overview of the Models for the CL Domain Transfer Experiments

3.2 Data Overview

The data and splits used in the domain transfer experiments were the same as Bourgeade et al., 2023. Overall, there are two topic-generic datasets, Davidson and Founta, four datasets that focus on misogyny and sexism, IberEval, Evalita,

HatEval_{women} and **Waseem_{sexism}**, and two datasets that focus on racism and xenophobia, **HatEval_{immigrant}** and **Waseem_{racism}**¹.

- **Davidson** (Davidson et al., 2017) contains tweets scraped by words and phrases identified as hate speech. The tweets are annotated as hate speech, offensive (but Not Hate speech) or Not Hate speech.
- The updated **Founta** (Founta et al., 2018) dataset by Kallumadi et al., 2020 contains tweets and the labels used for the annotation were offensive, abusive, Hate, aggressive, cyberbullying, spam or normal text.
- **IberEval** (Fersini, Rosso, & Anzovino, 2018) was also part of a shared task in misogyny identification and the tweets were collected in the same way as Evalita.
- **Evalita** (Fersini, Nozza, & Rosso, 2018), part of the Evalita 2018 campaign, focuses on misogyny identification and was collected via keyword searching, monitoring potential victims’ accounts, identifying misogynous accounts and downloading their history.
- **HatEval** (Basile et al., 2019), born out of a SemEval-2019 shared task, contains tweets, focuses in speech against immigrants and women and was collected in a similar fashion as Evalita and IberEval. **HatEval_{women}** represents the part of HatEval that was split with the misogynist subset. **HatEval_{immigrant}** represents the part of HatEval that was split with the anti-immigrant subset.
- **Waseem** (**Talat2016HateSO-waseem**) focuses on misogyny and racism and contains tweets collected through an initial manual search of slurs and Hate terms, as well as entities commonly associated with hate speech and tweets from prevalent accounts identified during that first initial search. The dataset was split by hate type, leaving us with **Waseem_{sexism}**, which represents the part of Waseem where the positive class are sexist messages, and **Waseem_{racism}**, which represents the part of Waseem that was split with the racist hate class. The samples of the negative class are the same for both sub-datasets.

Dataset	Hate Domain	Sample Size	Train/Val/Test Proportion	Balance Positive Class	Year of Publication
Davidson	General	5590	0.77/0.03/0.20	0.26/0.26/0.26	2017
Founta	General	57355	0.77/0.03/0.20	0.07/0.07/0.07	2018, updated in 2020
IberEval	Misogyny	3977	0.79/0.03/0.18	0.47/0.46/0.39	2018
Evalita	Misogyny	5000	0.77/0.03/0.20	0.45/0.45/0.46	2018
HatEval _{women}	Misogyny	6472	0.69/0.08/0.23	0.44/0.47/0.42	2019
Waseem _{sexism}	Sexism	14531	0.77/0.03/0.20	0.22/0.22/0.22	2016
HatEval _{immigrant}	Xenophobia	6499	0.69/0.08/0.23	0.40/0.39/0.42	2019
Waseem _{racism}	Racism	13272	0.77/0.03/0.20	0.15/0.15/0.15	2016

TABLE 3.2: Overview of the CL Domain Transfer Experiments Datasets

The data was processed by replacing user tags and urls by [USER] and [URL] using regular expressions as well as replacing emojis with their verbalization with the **emoji** Python library.

¹The color boxes represent the type of hate of the dataset: **general**, **misogyny**, **sexism**, **xenophobia** and **racism**.

3.2.1 Data Limitations

There are certain limitations and considerations to take into account when using these datasets. Bourgeade et al., 2023 already identified dataset compatibility issues as far as annotation inconsistencies, datasets that concern a specific type of hate potentially labeling an out-of-domain type of hate as part of the negative class, class imbalance and even conflicting definitions of hate speech and whether the datasets reflect that definition or not, an entire issue of its own studied in Khurana et al., 2025.

On top of these limitations, it is worth noting that IberEval and Evalita’s data collection process was very similar and performed during a close time period, 2017, so there are duplicates in their train splits that should be accounted for because, if both datasets were used for training in an experiment’s data stream, there would be unaccounted data replay taking place. Aside from that, there is also data contamination with respect to the training and testing splits of all the sexism and misogyny-focused datasets. The experimental data streams have been crafted so that the leakage is minimal whilst the experiments are still comparable to Bourgeade et al., 2023’s results and the degree of contamination will be highlighted for each experiment². As far as the Waseem sub-datasets, since they share their entire data for the negative class, there will be no data stream where we train on both. It is obvious that the performance of the negative class by a model trained on one of the sub-datasets will not be representative of the model’s capabilities. However, since the positive hate class represents a different hate domain with its own distinct samples in each sub-dataset’s positive class, we can account for the model’s performance in the positive class.

3.3 General Experimental Setting

Each of the Domain Transfer experiments consists of one datastream, of a certain known length, which a model will train through in a sequential way. For each dataset within the datastream the objective is the same: determine whether a message is hateful or not. This set up follows a standard Offline, Domain-Incremental Learning scenario: the task is known and it is the same for each dataset (binary classification), the possible classes remain the same across the stream (hateful or not hateful), we are interested in capturing the shifting data distributions across the datasets, and the broader domains they belong to, and we have access to all the training data for each training stage before it begins but not after.

For each experiment we will consider 7 kinds of models for each PLM, their training details can be seen in Table 3.4. The three baselines will be:

1. Merged Fine-tune (M-FT): sometimes referred to as i.i.d. (independent and identically distributed), consists of a single model for each PLM trained on the union of all the datasets we consider for each experiment. This model will address a would-be situation where we would be able to train our model in all our experiment data simultaneously instead of having to divide the process by dataset. This baseline will represent a more realistic upper-bound and each PLM will have one Merged Fine-tune per experiment.
2. Sequential Fine-tune (SEQ-FT): the PLM trained incrementally on the data stream without using Continual Learning techniques. This model should represent a lower-bound and a starting point from where Continual Learning techniques should improve the performance of the PLM.

²For a general discussion on the data contamination of the datasets, refer to Appendix B.

There will be four Continual Learning models ³ per experiment and PLM:

1. EWC: the PLM trained on the experiment’s data stream applying the supervised regularization technique Elastic Weight Consolidation.
2. MAS: the PLM trained on the experiment’s data stream applying the unsupervised regularization technique Memory Aware Synapses.
3. LwF: the PLM trained on the experiment’s data stream applying the knowledge distillation technique Learning without Forgetting.
4. A-GEM: the PLM trained on the experiment’s data stream applying the replay and optimization technique Averaged Gradient Episodic Memory.

Training Method	Data	Seq.	CL Method	CL Type	Experiment
M-FT	Union of the Data Stream	No	No	NA	Baseline
SEQ-FT	Data Stream	Yes	No	NA	CL Baseline
EWC	Data Stream	Yes	Yes	Regularization	CL Exp.
MAS	Data Stream	Yes	Yes	Regularization	CL Exp.
LwF	Data Stream	Yes	Yes	Distillation	CL Exp.
A-GEM	Data Stream	Yes	Yes	Replay and Gradient Projection	CL Exp.

TABLE 3.3: Type of PLM Models for each Experiment

Aside from the PLM-related baselines, we will also perform zero and few-shot experiments for each individual dataset with the LLMs discussed in 3.1⁴.

Training Method	Epochs	Batch Size	Learning Rate	Initial CL Hyperparameters
M-FT	8	32	1e-5	N/A
SEQ-FT	8 x number of tasks	32	1e-5	N/A
EWC	8 x number of tasks	32	1e-5	EWC LAMBDA=1500
MAS	8 x number of tasks	32	1e-5	MAS LAMBDA=1000
LwF	8 x number of tasks	32	1e-5	LwF LAMBDA=1, Temperature=2
A-GEM	8 x number of tasks	32	1e-5	Memory Size Proportion=2.5%

TABLE 3.4: PLM Models Training Settings

3.4 Continual Learning Experiments

Three data orders, each of different lengths, were considered to cover likely and realistic scenarios:

1. Training with data of alternating domains, switching between misogyny and racism and xenophobia (**Alt. Mis. & Rac.**), with a length of three.
2. Training on different datasets of the same hate speech domain and then changing to a new hate domain, going from xenophobia and racism to misogyny (**Rac. → Mis.**), with a length of four.
3. An initial training with general hate speech data, followed by further training on alternating hate speech domains, going from general into alternating misogyny and xenophobia and racism (**Gen. → Alt. Mis. & Rac.**), with a length of six.

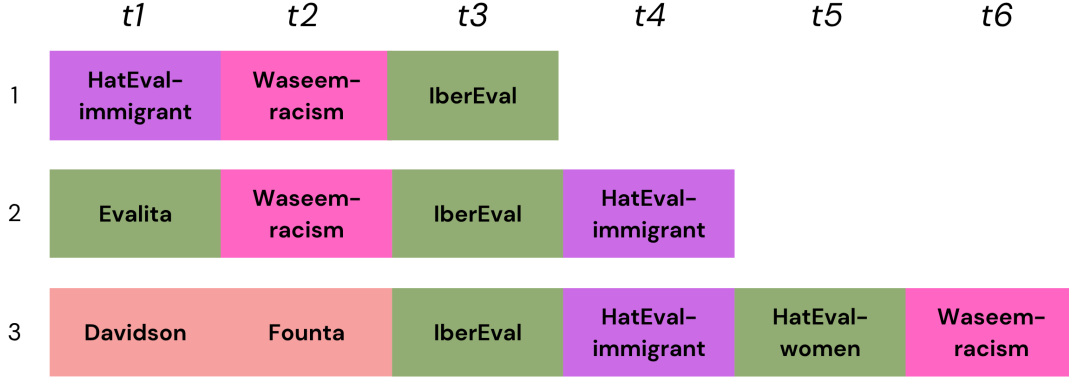


FIGURE 3.1: Overview of the Domain Transfer Experiment Streams

The results in terms of continual learning techniques will be presented first for all the experiments in general, without taking into account the zero shot performance on datasets that were not part of the experiment. Afterwards, each experiment will have its own dedicated section. Within each experiment’s section we will discuss the best continual learning model in more detail, reporting its performance over time as well as its zero shot capabilities in the zero shot datasets it was not trained with. It is worth noting that all the metrics have been calculated in terms of F1 score unless explicitly indicated otherwise.

3.4.1 General Continual Learning Results

PLM	CL Technique	CL Metrics				
		LAST	Average Incremental F1	Transfer	BWT	FWT
RoBERTa	A-GEM	0.74 ± 0.02	0.74 ± 0.12	0.74 ± 0.15	0.01 ± 0.05	0.33 ± 0.05
	EWC	0.66 ± 0.07	0.71 ± 0.06	0.67 ± 0.05	-0.10 ± 0.06	0.26 ± 0.08
	LwF	0.70 ± 0.06	0.71 ± 0.11	0.70 ± 0.13	-0.01 ± 0.07	0.31 ± 0.05
	MAS	0.67 ± 0.03	0.69 ± 0.12	0.68 ± 0.15	-0.01 ± 0.01	0.33 ± 0.05
	SEQ-FT	0.68 ± 0.07	0.71 ± 0.07	0.69 ± 0.05	-0.06 ± 0.08	0.27 ± 0.07
HateBERT	A-GEM	0.72 ± 0.02	0.70 ± 0.10	0.70 ± 0.10	0.02 ± 0.06	0.12 ± 0.04
	EWC	0.68 ± 0.04	0.69 ± 0.07	0.67 ± 0.07	-0.03 ± 0.05	0.09 ± 0.04
	LwF	0.68 ± 0.06	0.68 ± 0.10	0.68 ± 0.10	0 ± 0.06	0.14 ± 0.02
	MAS	0.63 ± 0.03	0.63 ± 0.12	0.64 ± 0.13	0.02 ± 0.03	0.20 ± 0.05
	SEQ-FT	0.69 ± 0.06	0.67 ± 0.08	0.64 ± 0.08	-0.01 ± 0.09	0.06 ± 0.02
fBERT	A-GEM	0.72 ± 0.01	0.72 ± 0.08	0.72 ± 0.10	0 ± 0.02	-0.07 ± 0.11
	EWC	0.68 ± 0.03	0.69 ± 0.07	0.67 ± 0.09	-0.01 ± 0.03	-0.11 ± 0.08
	LwF	0.69 ± 0.04	0.69 ± 0.08	0.69 ± 0.10	0.01 ± 0.04	-0.10 ± 0.08
	MAS	0.60 ± 0.01	0.64 ± 0.10	0.65 ± 0.12	0.01 ± 0.01	-0.07 ± 0.07
	SEQ-FT	0.70 ± 0.02	0.69 ± 0.07	0.67 ± 0.10	0 ± 0.02	-0.11 ± 0.08

TABLE 3.5: Continual Learning Results Aggregated.

Broadly looking at the results in 3.5, it is noticeable that, although an improvement, the average difference between the SEQ-FT baseline and the different continual learning techniques is not that high. The best performing continual learning models improve the baseline on a range of 2% to 5% in the metrics concerning the general

³For an introduction to each Continual Learning Technique, refer to A.1.1.

⁴The results are reported in Appendix D.

performance metrics (LAST, AI-F1 and Transfer) and the difference in terms of stability (BWT) and plasticity (FWT) follows a similar pattern. As will be seen later, part of this is due to the low performance of continual models in the Rac. $\rightarrow$ Mis. experiment. Another reason as to why this may be the case is that the initial hyperparameters were mostly set following foundational papers which were concerned with computer vision tasks and setting them to more strongly favour previous or future tasks may improve its performance.

In terms of general performance metrics, the best performing model is RoBERTa, specifically when paired with A-GEM. In contrast, it is noticeable that MAS performs much worse for HateBERT and fBERT. In terms of stability, there are not big differences and the performance greatly varies with respect to each experiment but HateBERT paired with MAS is the best model in that regard. This may seem odd at first given its poorer AI-F1 score but the fact that its F1 scores across the datastream are lower explain it: because the performance, albeit low, does not degrade as much, its stability is greater. As far as plasticity is concerned, RoBERTa with A-GEM or MAS are the best models and it is remarkable that with a higher plasticity, RoBERTa with A-GEM’s backward transfer remains positive on average, that is, not only does it learn quick, the speed of its learning also does not degrade its performance on previous datasets in the stream.

CL Technique	CL Metrics				
	LAST	Average Incremental F1	Transfer	BWT	FWT
A-GEM	0.73 ± 0.02	0.72 ± 0.09	0.72 ± 0.10	0.01 ± 0.04	0.13 ± 0.18
EWC	0.67 ± 0.05	0.69 ± 0.06	0.67 ± 0.06	-0.05 ± 0.06	0.08 ± 0.17
LwF	0.69 ± 0.05	0.69 ± 0.09	0.69 ± 0.10	0 ± 0.05	0.12 ± 0.19
MAS	0.63 ± 0.04	0.65 ± 0.10	0.66 ± 0.12	0 ± 0.02	0.15 ± 0.19
SEQ-FT	0.69 ± 0.05	0.69 ± 0.07	0.67 ± 0.07	-0.02 ± 0.07	0.07 ± 0.17

TABLE 3.6: Aggregated Results by Continual Learning Technique.

Table 3.6 corroborates the previous analysis regardless of the model used. A-GEM is, on average, the best in every continual learning metric except MAS’ forward transfer, which is explained by its lower AI-F1 score and broader range for improvement given its worse F1 scores.

Observing the aggregate continual learning metrics per experiment below, we can confirm an initial intuition that the longest experiment (Gen. $\rightarrow$ Alt. Mis. & Rac.) will have the worse LAST performance as introducing more datasets can lead to more instability across training stages, indicated by the backward transfer score, which is also the worst of the set of experiments. Keep in mind that the experiment’s length is different so each one learns a different number of tasks.

Experiment	CL Metrics				
	LAST	Average Incremental F1	Transfer	BWT	FWT
Rac. $\rightarrow$ Mis.	0.69 ± 0.05	0.59 ± 0.04	0.57 ± 0.04	0.03 ± 0.03	0.10 ± 0.21
Alt. Mis. & Rac.	0.70 ± 0.05	0.72 ± 0.05	0.71 ± 0.06	-0.01 ± 0.05	0.09 ± 0.16
Gen. $\rightarrow$ Alt. Mis. & Rac.	0.66 ± 0.04	0.76 ± 0.02	0.78 ± 0.02	-0.04 ± 0.04	0.17 ± 0.16

TABLE 3.7: Aggregated Results by Experiment excluding SEQ-FT.

As far as determining which model was more suited for continual learning in our setting, RoBERTa seems like a clear winner in the general and plasticity performance metrics although its backward transfer is worse due to the inability of RoBERTa with EwC to transfer knowledge from current tasks to previous ones. As can be seen in 3.5, the lambda parameter set for RoBERTa should be higher compared to the EwC lambda parameters of the other models in order to avoid such a high forgetting of old tasks.

PLM	CL Metrics				
	LAST	Average Incremental F1	Transfer	BWT	FWT
RoBERTa	0.69 ± 0.05	0.71 ± 0.09	0.70 ± 0.11	-0.03 ± 0.06	0.31 ± 0.06
HateBERT	0.68 ± 0.05	0.68 ± 0.08	0.67 ± 0.09	0.00 ± 0.06	0.13 ± 0.05
fBERT	0.67 ± 0.05	0.69 ± 0.07	0.68 ± 0.09	0.00 ± 0.02	-0.09 ± 0.08

TABLE 3.8: Aggregated Results by Model excluding SEQ-FT.

Lastly, it should be mentioned that the longest experiment does not yield the better overall results, even though it *sees* more tasks. The following is the average LAST performance for each experiment in every dataset, regardless of whether the experiment’s model was trained on the dataset or not (zero-shot):

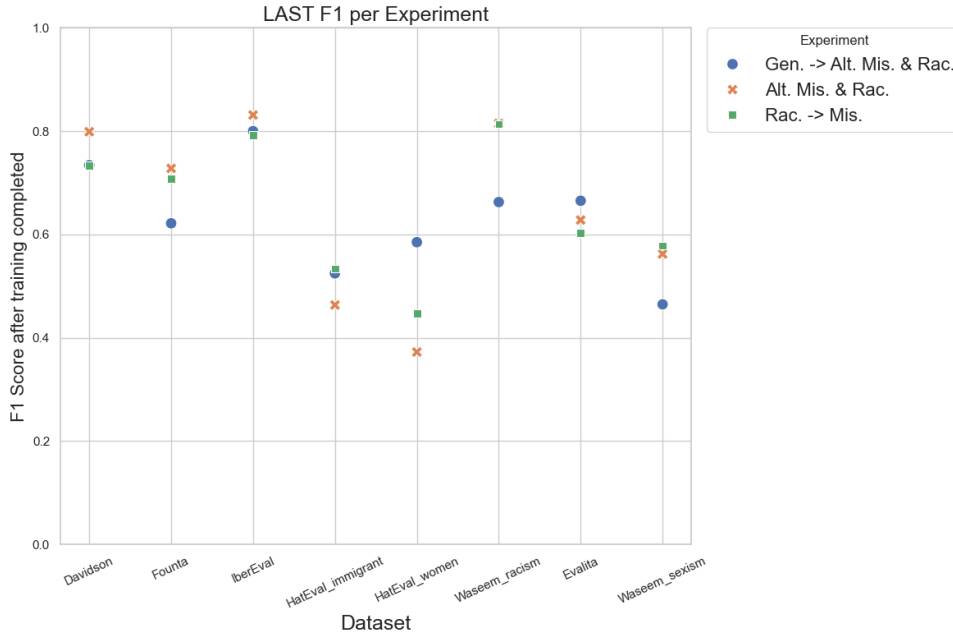


FIGURE 3.2: Aggregated LAST Performance by Experiment

We can observe in 3.2 that the experiment Alt. Mis. & Rac. leads in the two general datasets it was not trained on and the largest experiment only leads in two datasets out of the eight.

3.4.2 Fine-grained Results

3.4.2.1 Experiment Rac. $\rightarrow$ Mis.

Moving on to the individual experiments, Rac. $\rightarrow$ Mis. was the least successful overall. The overall best performing models were the SEQ-FT baselines of RoBERTa and HateBERT, and RoBERTa with A-GEM on the continual learning side. HateBERT SEQ-FT achieved the best backward transfer and RoBERTa with MAS the best forward transfer, although, as was discussed earlier, this is consistent with the fact that its overall performance is much worse. For the discussion of the experiments, we will focus on RoBERTa with A-GEM.

PLM	CL Technique	CL Metrics				
		LAST	Average Incremental F1	Transfer	BWT	FWT
RoBERTa	A-GEM	<u>0.75</u>	0.60	0.56	0.06	0.31
	EWC	0.71	<u>0.64</u>	0.63	-0.03	0.28
	LwF	0.71	0.58	0.56	0.03	0.33
	MAS	0.63	0.55	0.51	0	<u>0.34</u>
	SEQ-FT	<u>0.75</u>	0.63	<u>0.64</u>	0.03	0.28
HateBERT	A-GEM	0.74	0.58	0.59	0.09	0.07
	EWC	0.73	0.61	0.62	0.03	0.12
	LwF	0.66	0.56	0.57	0.06	0.16
	MAS	0.61	0.50	0.50	0.05	0.26
	SEQ-FT	<u>0.75</u>	0.58	0.56	<u>0.10</u>	0.04
fBERT	A-GEM	0.73	0.63	0.61	0.03	-0.17
	EWC	0.71	0.62	0.60	0.02	-0.17
	LwF	0.68	0.60	0.58	0.01	-0.17
	MAS	0.59	0.55	0.54	0.01	-0.14
	SEQ-FT	0.71	0.62	0.60	0.02	-0.17

TABLE 3.9: Experiment Alt. Mis. & Rac. Continual Learning Results

In order to assess the performance of the models, we report the M-FT upper bound baseline first. The M-FT baseline performs quite well in `Waseemracism` and `IberEval` but the performance of the hate class in `Waseemsexism` is extremely underwhelming. Our model improves the final performance on the `HatEvalimmigrant` dataset and matches `IberEval` while keeping very similar hate class scores. The SEQ-FT baseline improves on `HatEvalimmigrant` too. Something to take into account is that for the training of the M-FT baselines, the data was shuffled while in the sequential models the data came split by dataset, as we described earlier. This could have a pseudo meta-learning effect whereby the models are better than what is supposed to be the realistic upper bound. In terms of the zero shot performances, there are no major differences either except for the fact that SEQ-FT's performance on the `Waseemsexism` Hate class is much better.

Dataset	Testing Dataset	Hate Domain	M-FT	SEQ-FT	A-GEM
			F1 / Hate F1	F1 / Hate F1	F1 / Hate F1
Trained	HatEval _{immigrant}	Xenophobia	0.49 / 0.62	0.55 / 0.60	0.53 / 0.62
	Waseem _{racism}	Racism	0.85 / 0.74	0.79 / 0.64	0.84 / 0.73
	IberEval	Misogyny	0.88 / 0.85	0.89 / 0.87	0.88 / 0.85
Zero Shot	Evalita	Misogyny	0.62 / 0.60	0.61 / 0.60	0.60 / 0.56
	Davidson	General	0.75 / 0.60	0.79 / 0.67	0.77 / 0.63
	Founta	General	0.64 / 0.31	0.65 / 0.34	0.63 / 0.30
	HatEval _{women}	Misogyny	0.49 / 0.63	0.50 / 0.63	0.57 / 0.65
	Waseem _{sexism}	Sexism	0.50 / 0.14	0.65 / 0.42	0.49 / 0.12

TABLE 3.10: RoBERTa LAST Performance in Rac. → Mis.

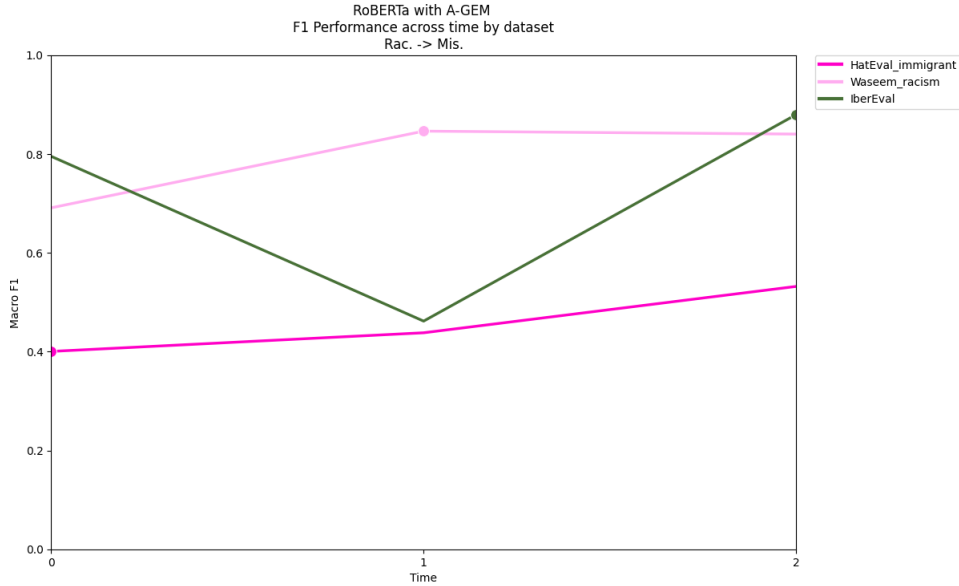


FIGURE 3.3: Rac. → Mis. RoBERTa F1 Performance Evolution

In terms of the training stream, the performance across time of `HatEvalimmigrant` and `Waseemracism` resembles what we would consider a good continual learning model, as it preserves its good performance across time and it even improves a little bit. Training in the second stage with `Waseemracism`, however, has a very detrimental effect in `IberEval`, possibly due to the out of domain hate type. It is interesting to see that the situation is not mirrored adversely, that is, training with `IberEval` does not have a severe negative effect on the rest of the tasks.

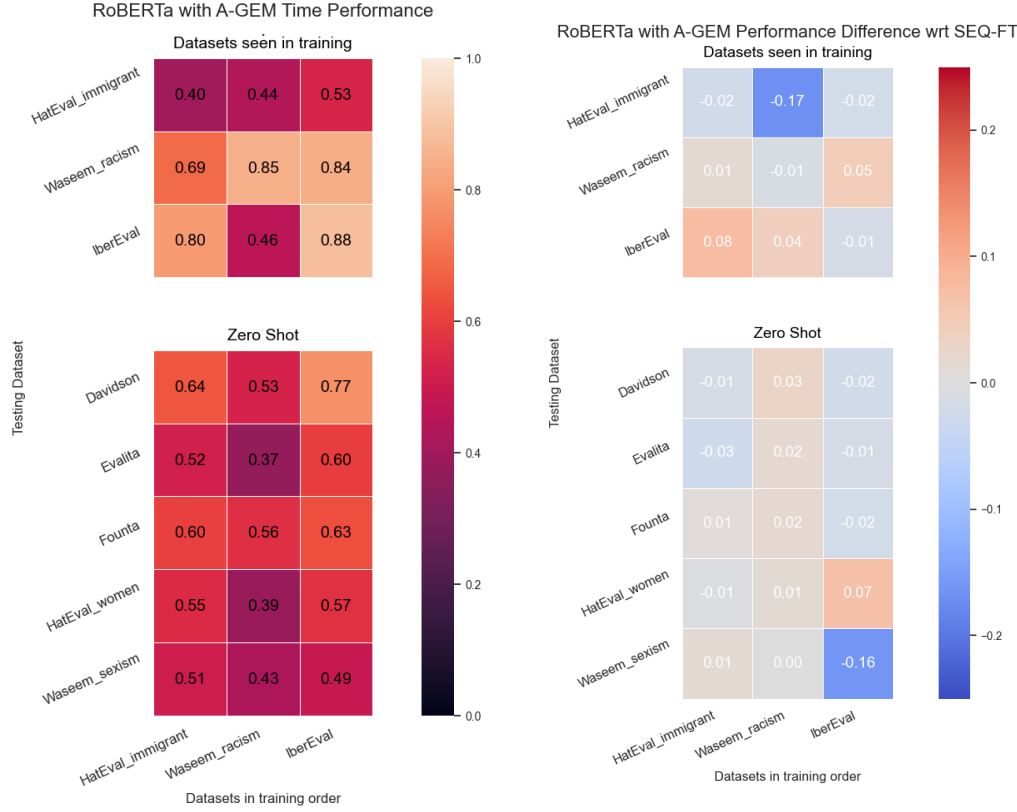


FIGURE 3.4: RoBERTa F1 Performance Evolution and Difference wrt SEQ-FT for Rac. → Mis.

Considering the difference with respect to the SEQ-FT, we can see that there is sort of an opposite effect as to what we mentioned previously. In the SEQ-FT, `HatEvalimmigrant` severely improves in the second stage before coming back down again at the end of the stream.

3.4.2.2 Experiment Alt. Mis. & Rac.

Generally speaking, the Alt. Mis. & Rac. experiment favored continual learning models more clearly. For RoBERTa, both A-GEM and LwF perform well and the latter even achieves positive backward transfer, that is, the performance of past tasks improves as new tasks are trained. This effect could be due to some sort of remembering the old tasks when the model is prompted with data from a domain it has already seen.

PLM	CL Technique	CL Metrics				
		LAST	Average Incremental F1	Transfer	BWT	FWT
RoBERTa	A-GEM	<u>0.76</u>	<u>0.82</u>	<u>0.84</u>	-0.02	<u>0.30</u>
	EWC	0.68	0.74	0.65	-0.12	0.17
	LwF	<u>0.76</u>	0.75	0.76	<u>0.04</u>	0.26
	MAS	0.68	0.72	0.73	-0.02	0.27
	SEQ-FT	0.70	0.75	0.67	-0.09	0.19
HateBERT	A-GEM	0.71	0.76	0.75	-0.03	0.15
	EWC	0.68	0.71	0.65	-0.07	0.04
	LwF	0.75	0.73	0.71	0.02	0.13
	MAS	0.63	0.66	0.67	0.01	0.16
	SEQ-FT	0.68	0.71	0.64	-0.06	0.05
fBERT	A-GEM	0.72	0.75	0.76	0.00	-0.07
	EWC	0.68	0.69	0.64	-0.02	-0.15
	LwF	0.73	0.71	0.71	<u>0.04</u>	-0.12
	MAS	0.60	0.63	0.65	0.02	-0.09
	SEQ-FT	0.68	0.69	0.64	-0.02	-0.15

TABLE 3.11: Experiment Gen. $\rightarrow$ Alt. Mis. & Rac. Continual Learning Results

In terms of the performance of RoBERTa A-GEM with respect to the baselines, the model outperforms the M-FT in all datasets aside from `HatEvalimmigrant` and, regarding zero shot tasks, `Davidson` and `Founta`. Compared to the SEQ-FT, the performance of the trained datasets is much better across the board but the continual learner failed to compete when it comes to zero shot performance against SEQ-FT.

Dataset	Testing Dataset	Hate Domain	M-FT		SEQ-FT		A-GEM	
			F1	Hate F1	F1	Hate F1	F1	Hate F1
Trained	Evalita	Misogyny	0.70	0.69	0.63	0.71	0.76	0.76
	Waseem _{racism}	Racism	0.84	0.73	0.80	0.67	0.84	0.73
	IberEval	Misogyny	0.88	0.84	0.92	0.91	0.95	0.94
	HatEval _{immigrant}	Xenophobia	0.52	0.62	0.44	0.62	0.48	0.62
Zero Shot	Davidson	General	0.73	0.56	0.83	0.75	0.65	0.43
	Founta	General	0.64	0.32	0.70	0.45	0.62	0.28
	HatEval _{women}	Misogyny	0.51	0.64	0.37	0.61	0.56	0.65
	Waseem _{sexism}	Sexism	0.50	0.13	0.68	0.49	0.60	0.34

TABLE 3.12: RoBERTa LAST Performance in Alt. Mis. & Rac.

As far as the evolution of the performance across training, it is remarkable that when training on `Waseemracism`, its performance in the dataset skyrockets but the out of domain tasks do not suffer as a consequence. In the last step, both racism and xenophobic datasets decrease a bit while `IberEval` increases, but that may be a byproduct of projecting the `IberEval` out of domain gradients. Overall, the shape of the performance from the first to the second time suggests that the datasets are compatible when trained on `Waseemracism` with the `Evalita` memory buffer.

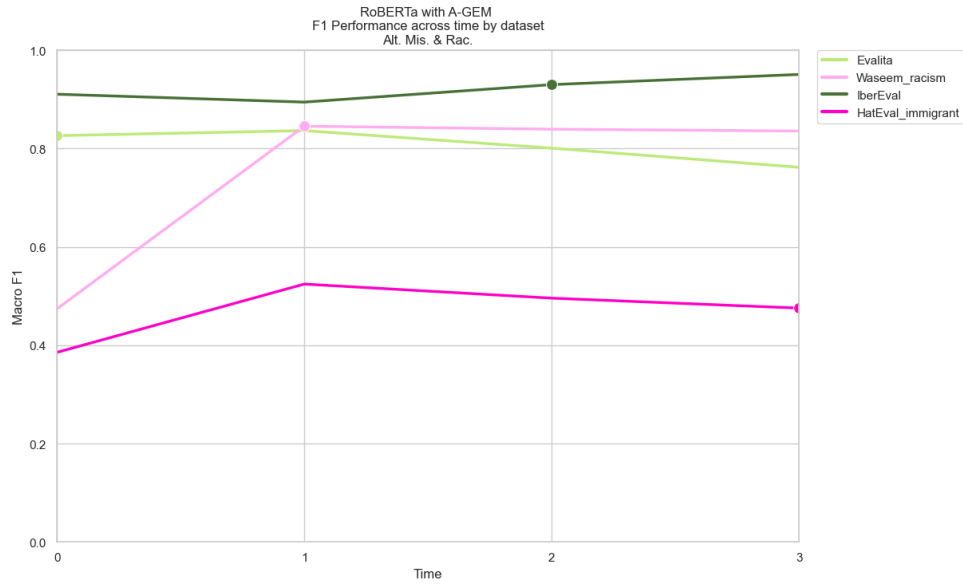


FIGURE 3.5: Alt. Mis. & Rac. RoBERTa F1 Performance Evolution

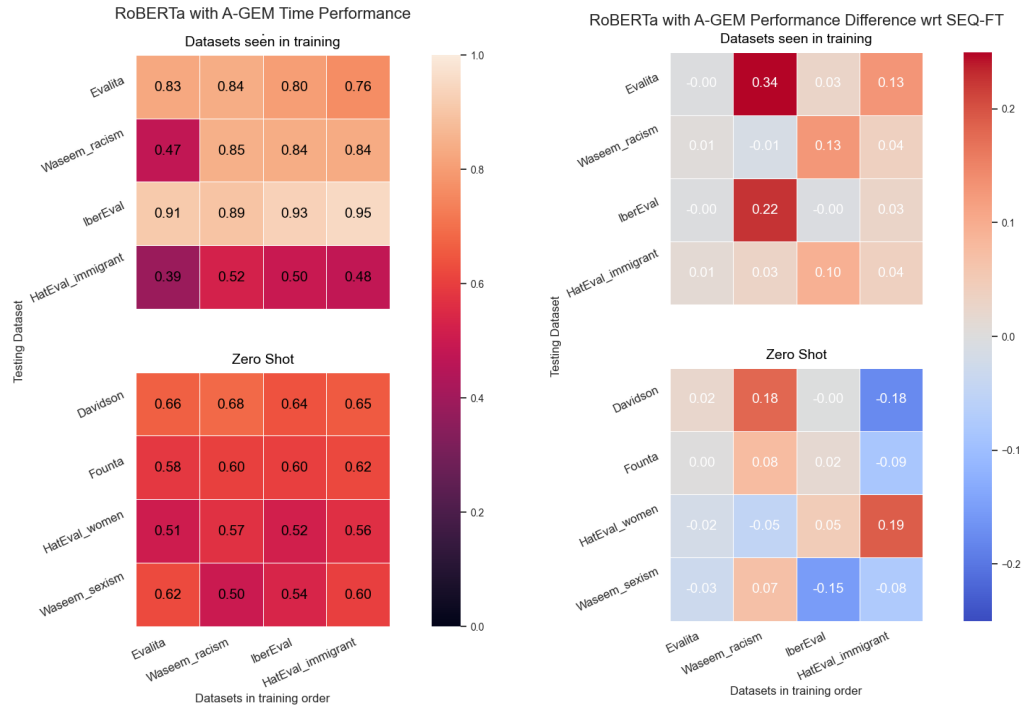


FIGURE 3.6: RoBERTa F1 Performance Evolution and Difference wrt SEQ-FT for Alt. Mis. & Rac.

Comparing the training with the SEQ-FT, the continual learner is always ahead in performance for the tasks in the datastream although as we discussed before, the model does struggle with zero shot performance at the end of the stream when the domain of the training data is xenophobia and the memory keeps misogynistic data to project the gradients.

3.4.2.3 Experiment Gen. $\rightarrow$ Alt. Mis. & Rac.

Finally, for the last experiment, it is interesting that the continual models of HateBERT and RoBERTa consistently outperform their SEQ-FT baselines although fBERT’s SEQ-FT is close to the best performing continual learning model. One limitation we are facing is that the hate specific models may have seen some of this data already and that could be why fBERT performs so well in this particular experiment despite the other baselines usually having higher LAST scores.

PLM	CL Technique	CL Metrics				
		LAST	Average Incremental F1	Transfer	BWT	FWT
RoBERTa	A-GEM	<u>0.72</u>	<u>0.81</u>	<u>0.82</u>	-0.02	<u>0.38</u>
	EWC	0.58	0.74	0.73	-0.14	0.33
	LwF	0.65	0.79	0.79	-0.09	0.35
	MAS	0.69	0.79	0.81	-0.01	<u>0.38</u>
	SEQ-FT	0.61	0.75	0.74	-0.12	0.33
HateBERT	A-GEM	0.71	0.76	0.77	<u>0</u>	0.12
	EWC	0.64	0.74	0.74	-0.04	0.10
	LwF	0.63	0.75	0.77	-0.06	0.12
	MAS	0.66	0.74	0.76	<u>0</u>	0.18
	SEQ-FT	0.63	0.72	0.71	-0.07	0.09
fBERT	A-GEM	0.71	0.78	0.79	-0.01	0.04
	EWC	0.65	0.76	0.77	-0.04	-0.02
	LwF	0.65	0.76	0.78	-0.04	-0.01
	MAS	0.62	0.75	0.77	<u>0</u>	0
	SEQ-FT	0.69	0.77	0.78	-0.01	-0.02

TABLE 3.13: Experiment Gen. $\rightarrow$ Alt. Mis. & Rac. Continual Learning Results

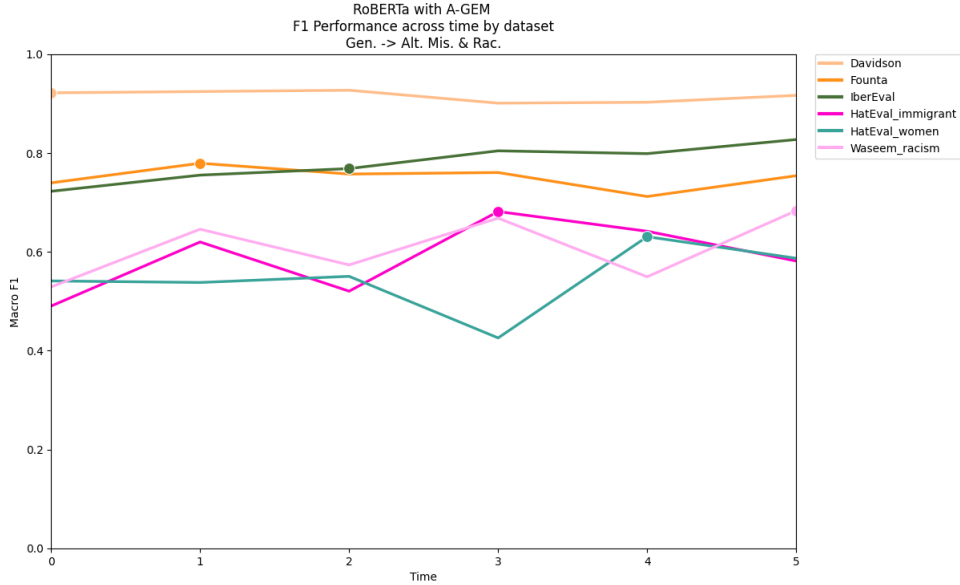
This experiment puts forward RoBERTa as quick learner as its forward transfer scores are significantly higher than the other PLMs and the other experiments. It is shocking how much RoBERTa EWC forgets while also not being able to be among the best models in terms of forward transfer; its regularization is so loose that there is a consistent loss in performance but at the same time not adequate enough to learn new tasks quicker.

In this experiment, M-FT is a stronger model and A-GEM is beaten in three out of the six in-stream datasets; the case of Waseem_{racism} and its hate class performance is very noticeable. However, SEQ-FT is much weaker, specially in its hate class scores.

Dataset Position	Testing Dataset	Hate Domain	M-FT		SEQ-FT		A-GEM	
			F1	Hate F1	F1	Hate F1	F1	Hate F1
Trained	Davidson	General	0.91	0.87	0.75	0.59	0.92	0.87
	Founta	General	0.77	0.56	0.56	0.16	0.75	0.54
	IberEval	Misogyny	0.86	0.82	0.77	0.67	0.83	0.77
	HatEval _{immigrant}	Xenophobia	0.58	0.64	0.42	0.11	0.58	0.64
	HatEval _{women}	Misogyny	0.51	0.64	0.65	0.66	0.59	0.65
	Waseem _{racism}	Racism	0.82	0.69	0.51	0.09	0.68	0.44
Zero Shot	Evalita	Misogyny	0.62	0.63	0.62	0.50	0.62	0.62
	Waseem _{sexism}	Sexism	0.53	0.20	0.46	0.04	0.51	0.16

TABLE 3.14: RoBERTa LAST Performance in Gen. $\rightarrow$ Alt. Mis. & Rac.

When it comes to the evolution of the performance, training on the general datasets first improves every other task performance. We can observe a sort of ping-pong effect in the racism and xenophobia datasets, as well as `HatEvalwomen` when trained in out of domain data. Paradoxically, the performance of `HatEvalimmigrant` does not bounce back at the last training stage despite training in a similar domain. Unfortunately, the performance of the general tasks degrades when the model trains on `HatEvalimmigrant`; training on general datasets improves every other domain specific dataset but that effect is not fully mirrored.

FIGURE 3.7: Gen. $\rightarrow$ Alt. Mis. & Rac. RoBERTa F1 Performance Evolution

In comparison with the SEQ-FT, most of the performance across time is similar during the first four stages and after training on `HatEvalwomen`, the continual model distances itself from SEQ-FT clearly. It may seem counter-intuitive because the performance overall goes down for the continual model but the SEQ-FT takes a much bigger loss, which suggests that `HatEvalwomen` at that point is not beneficial for the model's performance.

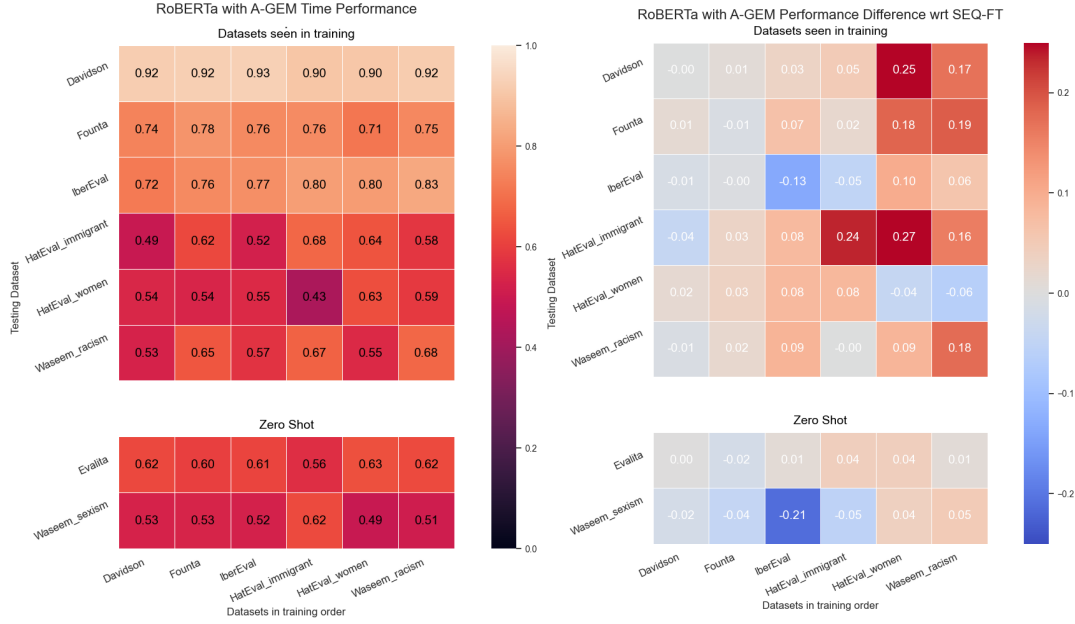


FIGURE 3.8: RoBERTa F1 Performance Evolution and Difference wrt SEQ-FT for Gen. $\rightarrow$ Alt. Mis. & Rac.

3.5 Prediction Sequences

After assessing how different continual learning techniques affect the performance on different datasets and experiments, now we turn our attention to how predictions change across the datastream while the models are incrementally trained on new data, and we will perform a qualitative analysis of the models' predictions at different points of the stream. We will evaluate both aspects for our RoBERTa A-GEM models trained with a memory buffer of 25% of the current data size at each time of the datastream.

When speaking of sequences of predictions, we will refer to **correct** or **incorrect** predictions. That is, for each experiment, we have a sequence which is as long as the experiment and whose values are binary, correct or incorrect, based on whether the label predicted for the model was correct or not.

Bare in mind that we performed zero-shot testing of datasets outside of the training datastream, however, for this analysis of prediction sequences we will only consider those data points that belonged to datasets part of the stream. Since the hate and not hate classes are not balanced in most datasets, we will consider the correct or incorrect prediction sequences for each class separately after a general discussion.

3.5.1 Prediction Sequences

In order to assess the way our models' predictions evolve, we will focus on three aspects:

- The general groups of prediction sequences, represented by the most representative clusters of sequences of each experiment regardless of class ⁵. Even though prediction sequences will be clustered under prototypes (medoids) that do not match them exactly, we aim to get a broad sense of how the predictions evolve

⁵For details on the clustering implementation, refer to [F](#).

in our models since examining each unique sequence would be too narrow of a focus.

- The most common sequences of predictions per class, since there may be significant differences as to how the models predict the hate and not hate classes.
- The points in the stream where predictions shift from correct to incorrect (and vice versa), which we will call **turns**. At this point, we will be more concerned with what happens in the middle of the model regardless of the final prediction being correct or incorrect. When we speak about turning predictions from correct to incorrect, we are ignoring those predictions which stay consistent so analysis does not take into account whether training at a specific time leads to very robust predictions.

3.5.1.1 Prediction Sequences for Experiment Rac. $\rightarrow$ Mis.

Overall, the majority of predictions were clustered under the Correct $\rightarrow$ Correct $\rightarrow$ Correct medoid and around 16% fell under the medoid for an ever-incorrect prediction.

Time 1 HatEval_{immigrant}	Time 2 Waseem_{racism}	Time 3 IberEval	Proportion
Incorrect	Incorrect	Incorrect	0.16
		Correct	0.11
Correct	Correct	Correct	0.68
		Incorrect	0.04

TABLE 3.15: Cluster Medoids for Rac. $\rightarrow$ Mis.

[E.1](#) indicates that the majority of hateful messages were correctly classified from the very beginning of the stream and continued to be correct throughout the experiment. Around 10% of the data points were never correctly classified and by training with IberEval at the end, around 18% of the data points' predictions were corrected. For the not hateful messages [E.2](#), we observe a similar pattern: most data points were always correctly classified and around 18% were never correctly classified.

Overall, [3.9](#) tells us 67% of prediction turns happened from incorrect to correct and, out of the correct turns, IberEval was responsible for 56% of them.

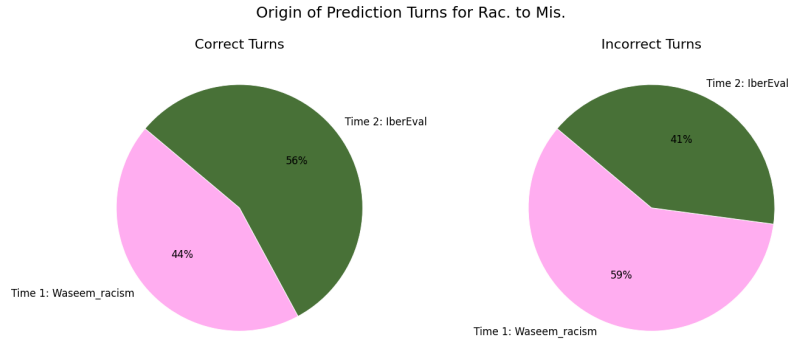


FIGURE 3.9: Prediction Turns for Rac. to Mis.

3.5.1.2 Prediction Sequences for Experiment Alt. Mis. & Rac.

Unsurprisingly, the majority of the predictions for Alt. Mis. & Rac. also fell under the always correct medoid. We also remark that we see a flip-flopping pattern emerge in this experiment, which is expected as this experiment was longer than the previous one.

Time 1	Time 2	Time 3	Time 4	Proportion
Evalita	Waseem _{racism}	IberEval	HatEval _{immigrant}	
Incorrect	Incorrect	Incorrect	Incorrect	0.05
			Correct	0.18
Correct	Correct	Correct	Correct	0.61
	Incorrect	Correct	Incorrect	0.16

TABLE 3.16: Cluster Medoids for Alt. Mis. & Rac.

E.3 also followed a similar pattern to the previous experiment’s per label sequences, with the main difference being that 29% of the hate class predictions were corrected at the final time after being incorrectly classified during most of the stream. As far as not hateful messages **E.2**, most of them were correctly classified across the datastream, but in this case, 18% of the predictions switched to incorrect predictions at the end.

In this experiment the distribution of the turns was more balanced, with 52% of the turns going from incorrect to correct. **HatEval_{immigrant}** stands out as the most unstable point: it was the task responsible for the most correct turns but also the most incorrect ones. **IberEval** did not have much influence overall although its effect could be described as slightly negative when we speak about turning predictions.

Once again, since we are only speaking about *turning predictions*, this does not mean that **IberEval** had a negative effect since predictions that were consistent are not taken account in **3.10**.

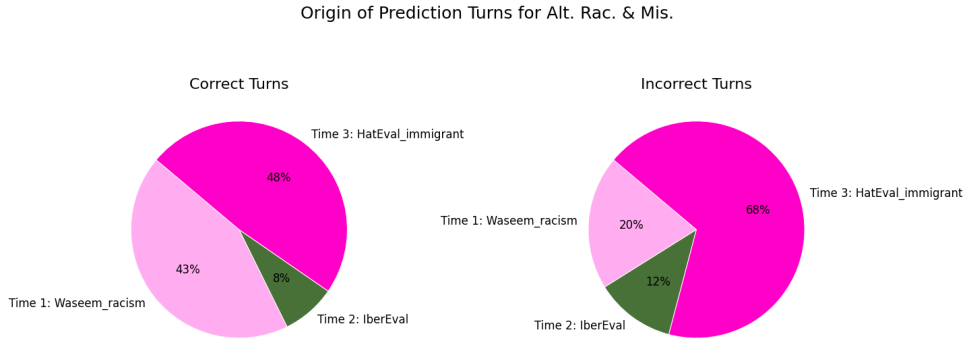


FIGURE 3.10: Prediction Turns for Alt. Rac. & Mis.

3.5.1.3 Prediction Sequences for Experiment Gen. → Alt. Mis. & Rac.

The most common medoid for Gen. → Alt. Mis. & Rac. was also the always correct sequence and we observe flip-flopping patterns in a much lesser capacity.

Time 1 Davidson	Time 2 Founta	Time 3 IberEval	Time 4 HatEval_{immigrant}	Time 5 HatEval_{women}	Time 6 Waseem_{racism}	Proportion
Incorrect	Incorrect	Incorrect	Incorrect	Incorrect	Incorrect	0.10
	Correct	Correct	Incorrect	Correct	Correct	0.02
		Correct	Correct	Correct	Correct	0.02
Correct	Correct	Correct	Correct	Correct	Correct	0.80
		Incorrect	Correct	Incorrect	Incorrect	0.04
	Incorrect	Correct	Incorrect	Correct	Correct	0.02

TABLE 3.17: Cluster Medoids for Experiment Gen. $\rightarrow$ Alt. Mis. & Rac.

As expected, there is more variety in terms of prediction patterns as the stream is longer, but the same trends continue for the hateful messages E.5. As in previous experiments, we observe that most predictions were either always correctly classified or always incorrectly classified. It is also noticeable that around 9% of the predictions went from firstly being misclassified until time 4-5 and then switching to a correct classification that was maintained until the end of the stream.

The not hateful message prediction sequences E.6 were dominated by an always correct behavior, an effect that we may attribute to Davidson and Founta being the largest datasets, and the latter having a larger proportion of Not Hate messages. Intuitively, we could say that the model learned to predict not hateful messages early in the stream with respect to the other experiments.

For this final experiment, 51% of the turns were correct. The beginning and ending of the stream were balanced in terms of turning predictions but the middle (HatEval_{immigrant} and HatEval_{women}) proved to be a bit more unstable.

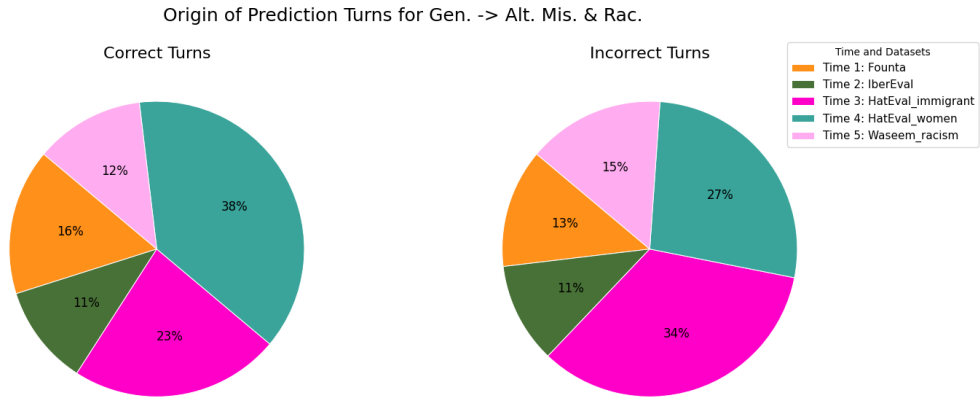


FIGURE 3.11: Prediction Turns for Gen. $\rightarrow$ Alt. Rac. & Mis.

3.5.2 Qualitative Analysis

3.5.2.1 Prediction of Vague Sentences

Following Bourgeade et al., 2023, we evaluate the predictions of our best model for each experiment (RoBERTa A-GEM) for all experiments with ambiguous sentences, sentences which lack context and sentences that are implicitly hate. The sentences are the following:

1. *What pronoun should we use to refer to people who are transitioning into mar-tians?*
2. *Women are intellectually deficient, islam says in sahih bukhari 1:6:301. disagree? you are islamophobe.*
3. *Oh honey. just oh, honey. bless your heart.*
4. *Maybe time to oil up the deer rifles, the ones with the hd night scopes to keep the "vermin" population down.*

Experiment	Sentence ID	Prediction	Predicted Probability
Gen. → Alt. Mis. & Rac.	1	Not Hate	0.94
Alt. Mis. & Rac.	1	Not Hate	0.98
Rac. → Mis.	1	Not Hate	0.98
Gen. → Alt. Mis. & Rac.	2	Hate	0.55
Alt. Mis. & Rac.	2	Hate	0.87
Rac. → Mis.	2	Hate	0.64
Gen. → Alt. Mis. & Rac.	3	Not Hate	0.93
Alt. Mis. & Rac.	3	Not Hate	0.96
Rac. → Mis.	3	Not Hate	0.98
Gen. → Alt. Mis. & Rac.	4	Not Hate	0.86
Alt. Mis. & Rac.	4	Hate	0.68
Rac. → Mis.	4	Not Hate	0.70

TABLE 3.18: Result of the Prediction of Vague Sentences

Taking the probability of the predicted class as a proxy for the model’s confidence, we would expect a perfect model to have little confidence on the predicted class of vague sentences. 3.18 shows that it is not the case for most of our cases. The first sentence, which could be described as ironically attacking transgender people, is very confidently predicted as not hateful.

The forth sentence is particularly interesting. The author of the tweet implicitly refers to a group of people as a vermin, a plague. It is the only sentence where the models disagree on the final prediction, with the Alt. Mis. & Rac. experiment model being the only one which assigned a hateful label. This result underscores both the importance of order and the effects of continual learning: training with the IberEval dataset at the last step with Waseem_{racism} in the memory buffer produces a wrong prediction while training with HatEval_{immigrant} and an IberEval buffer produces a correct prediction.

3.5.2.2 Error Analysis

Since we not only care about the final prediction of a model but also how their predictions evolve, we have selected sentences common to every experiment’s test set for a time-based analysis.

The sentences are the following:

1. *[USER] those are waaaaay more spendy. this is the drive i have. just considering using built-in or if i should go software.*
2. *23. Use a combination of the following phrases at least three dozens times a day on social media: cuck, soy, soyboy, SJW, beta male, mangina, snowflake, libtard, MAGA, 14/88, deus vult, dindu, rapefugee, kek. #Tradlife*
3. *Wow they tried to give me a FEMALE DOCTOR TODAY. I didn't even know those EXISTED...*
4. *[USER] #BuildTheDamnWall #BuildThatDamnWall #BuildThatWall #BuildTheWall #4MollieTibbetts #BuildTheWallNow Need I say more? Solves ALL the problems!!*

Sentence	Difficulty	Hate Domain	Label
1	Clearly Not Hateful	NA	Not Hate
2	Ambiguous	NA	Not Hate
3	Clearly Hateful	Sexist	Hate
4	Clearly Hateful	Anti-Immigration	Hate

TABLE 3.19: Test Sentences Information

As expected, the prediction for the not hateful sentence remained consistent and correct in each step in every experiment 3.20, something that we can explain by the fact that the negative class represents the same category in most datasets⁶.

TABLE 3.20: Results for Sentence 1: Clearly Not Hateful

Experiment	Time	Dataset at time t	Prediction	Pred. Probability	Result
Gen. → Alt. Mis. & Rac.	1	Davidson	Not Hate	0.99	Correct
	2	Founta	Not Hate	1.00	Correct
	3	IberEval	Not Hate	0.96	Correct
	4	HatEval _{immigrant}	Not Hate	0.88	Correct
	5	HatEval _{women}	Not Hate	0.99	Correct
	6	Waseem _{racism}	Not Hate	0.99	Correct
Alt. Mis. & Rac.	1	Evalita	Not Hate	0.97	Correct
	2	Waseem _{racism}	Not Hate	0.99	Correct
	3	IberEval	Not Hate	0.99	Correct
	4	HatEval _{immigrant}	Not Hate	0.99	Correct
Rac. → Mis.	1	HatEval _{immigrant}	Not Hate	0.95	Correct
	2	Waseem _{racism}	Not Hate	0.99	Correct
	3	IberEval	Not Hate	0.99	Correct

⁶Except those exceptions outlined in 3.2.1

Given that the context of the second tweet is missing, we can only infer that the message was part of a thread where this particular tweet is the 23rd tweet. The ambiguous nature of the message comes from the fact that the tweeter seems to be given some sort of instructions to, perhaps, tweet like a hateful user and they are using an ironic tone. The prediction for each model’s experiment does not change within each experiment 3.21 although the probability of the predicted class in Gen. → Alt. Mis. & Rac. and Alt. Mis. & Rac. gradually decreases. Because most of the hateful words used in the message relate to the domains of politics and immigration, it is logical that the experiment, which includes half of its training data related to misogyny and sexism, does not produce a correct prediction although this does not necessarily mean that the model understood the ironic tone.

TABLE 3.21: Results for Sentence 2: Ambiguous

Experiment	Time	Dataset at time t	Prediction	Pred. Probability	Result
Gen. → Alt. Mis. & Rac.	1	Davidson	Hate	0.91	Incorrect
	2	Founta	Hate	1.00	Incorrect
	3	IberEval	Hate	0.78	Incorrect
	4	HatEval _{immigrant}	Hate	0.85	Incorrect
	5	HatEval _{women}	Hate	0.69	Incorrect
	6	Waseem _{racism}	Hate	0.75	Incorrect
Alt. Mis. & Rac.	1	Evalita	Not Hate	0.85	Correct
	2	Waseem _{racism}	Not Hate	0.81	Correct
	3	IberEval	Not Hate	0.88	Correct
	4	HatEval _{immigrant}	Not Hate	0.50	Correct
Rac. → Mis.	1	HatEval _{immigrant}	Hate	0.76	Incorrect
	2	Waseem _{racism}	Hate	0.71	Incorrect
	3	IberEval	Hate	0.84	Incorrect

The prediction evolution for the clearly sexist message follows the cluster medoids we discussed in 3.5.1: either the prediction does not change, changes at the end or shifts back and forth during the middle of the longest experiment. Although both Alt. Mis. & Rac. got the sentence right from the beginning and Rac. → Mis. switched to a correct prediction after trianing on the domain of the hateful message, Gem. → Alt. only predicted the tweet correctly during time 4, after training with HatEval_{immigrant} with IberEval in the memory buffer.

TABLE 3.22: Results for Sentence 3: Sexist

Experiment	Time	Dataset at time t	Prediction	Pred. Probability	Result
Gen. → Alt. Mis. & Rac.	1	Davidson	Not Hate	0.80	Incorrect
	2	Founta	Not Hate	1.00	Incorrect
	3	IberEval	Not Hate	0.67	Incorrect
	4	HatEval _{immigrant}	Hate	0.52	Correct
	5	HatEval _{women}	Not Hate	0.70	Incorrect
	6	Waseem _{racism}	Not Hate	0.60	Incorrect
Alt. Mis. & Rac.	1	Evalita	Hate	0.79	Correct
	2	Waseem _{racism}	Hate	0.90	Correct
	3	IberEval	Hate	0.69	Correct
	4	HatEval _{immigrant}	Hate	0.94	Correct
Rac. → Mis.	1	HatEval _{immigrant}	Not Hate	0.53	Incorrect
	2	Waseem _{racism}	Not Hate	0.82	Incorrect
	3	IberEval	Hate	0.61	Correct

The patterns we observe for the final anti-immigration message are almost identical to those of the sexist message. In the two shorter experiments, the prediction switches to a correct one after training with in-domain data while the longer experiment never predicts it correctly. The influence of HatEval_{immigrant} is clear in the two shorter experiments as hashtags related to the building of the wall between Mexico and the US are prevalent there.

TABLE 3.23: Results for Sentence 4: Anti-Immigrant

Experiment	Time	Dataset at time t	Prediction	Pred. Probability	Result
Gen. → Alt. Mis. & Rac.	1	Davidson	Not Hate	0.98	Incorrect
	2	Founta	Not Hate	1.00	Incorrect
	3	IberEval	Not Hate	0.97	Incorrect
	4	HatEval _{immigrant}	Not Hate	0.83	Incorrect
	5	HatEval _{women}	Not Hate	0.74	Incorrect
	6	Waseem _{racism}	Not Hate	0.71	Incorrect
Alt. Mis. & Rac.	1	Evalita	Not Hate	0.91	Incorrect
	2	Waseem _{racism}	Not Hate	0.98	Incorrect
	3	IberEval	Not Hate	0.98	Incorrect
	4	HatEval _{immigrant}	Hate	0.97	Correct
Rac. → Mis.	1	HatEval _{immigrant}	Hate	0.96	Correct
	2	Waseem _{racism}	Hate	0.97	Correct
	3	IberEval	Hate	0.84	Correct

3.6 Hyperparameter Sensitivity

Unequivocally, A-GEM was the better continual learning technique for our experiments. Since its gradient projection depends on the samples that are replayed, it is in a model’s best interest to keep the memory stored as low as possible and save the dataset’s more representative samples in memory to avoid unnecessary computations. For the standard configuration we randomly selected 2.5% of the current training dataset to replay during the immediately following task and the memory

was reset after each training stage. In order to assess the trade-off between amount of data stored and increase in performance, the experiments were repeated with the best performing mode, RoBERTa, with a memory proportion of 1% and 5% (except for the Gen. $\rightarrow$ Alt. Mis. & Rac. experiment).

Model	CL Technique	Hyperparams.	Experiments		
			Gen. $\rightarrow$ Alt. Mis. & Rac.	Alt. Mis. & Rac.	Rac. $\rightarrow$ Mis.
RoBERTa	SEQ-FT	None	0.61	0.70	<u>0.75</u>
	A-GEM	% of train data in Memory: 1%	<u>0.72</u>	0.74	0.74
		% of train data in Memory: 2.5%	<u>0.72</u>	<u>0.76</u>	<u>0.75</u>
		% of train data in Memory: 5%	N/A	<u>0.76</u>	<u>0.75</u>

TABLE 3.24: RoBERTa LAST A-GEM Performance with different Hyperparams.

Decreasing the memory size to just 1% did not affect the performance of RoBERTa in a major way: the performance for the longest experiment was the same and the other experiments' performance only decreased 1% and 2%.

To determine how beneficial it is to input more memory, we first repeat the Alt. Mis. & and Rac. Rac. $\rightarrow$ Mis. experiments. Given that the performance did not increase, we did not perform the Gen. $\rightarrow$ Alt. Mis. & Rac. experiment with more memory due to how much more computationally intensive it is.

In terms of the change in the performance of the hate class, the gains were also minimal; decreasing the memory to 1% only decrease the performance by 2% in each experiment and increasing the memory to 5% only yielded a 1% increase in the Rac. $\rightarrow$ Mis. experiment.

Model	CL Technique	Hyperparams.	Experiments		
			Gen. $\rightarrow$ Alt. Mis. & Rac.	Alt. Mis. & Rac.	Rac. $\rightarrow$ Mis.
RoBERTa	SEQ-FT	None	0.38	0.73	0.70
	A-GEM	% of train data kept in Memory: 1%	0.63	0.74	0.71
		% of train data kept in Memory: 2.5%	<u>0.65</u>	<u>0.76</u>	0.73
		% of train data kept in Memory: 5%	N/A	<u>0.76</u>	<u>0.74</u>

TABLE 3.25: RoBERTa Avg. Hate F1 Performance with different Hyperparams.

What these hyperparameter search results underscore is the fact that because we are projecting the gradients instead of explicitly replaying past batches, the memory size is not as important as those samples in the memory being the most representative of the previous task. That is, there is an inherent ceiling to how much the performance improves when the sampling of the samples is random.

3.7 Hate Class Predictions

It is crucial that Hate Speech Detection models perform well detecting the hateful messages. Motivated by the initial hypothesis that the regularization, replay and distillation characteristic of continual learning techniques may, perhaps inadvertently, improve hate-specific predictions compared to sequential finetuning, we further report the difference in F1 Hate class performance with respect to the SEQ-FT baseline for every pair of model, technique and experiment.

Experiment	Model	CL Techniques			
		A-GEM	EWC	LwF	MAS
Gen. $\rightarrow$ Alt. Mis. & Rac.	RoBERTa	196.65	-31.81	31.03	127.28
	HateBERT	251.31	-2.19	-1.64	69.93
	fBERT	16.68	-5.31	-11.34	-3.82
Alt. Mis. & Rac.	RoBERTa	4.93	-2.58	-10.26	-22.29
	HateBERT	2.04	-0.57	-2.89	-41.43
	fBERT	4.28	-0.02	-4.21	-40.83
Rac. $\rightarrow$ Mis.	RoBERTa	4.80	-2.94	-5.99	-11.13
	HateBERT	0.84	0.72	-14.65	-21.42
	fBERT	1.01	0.08	-9.58	-25.15

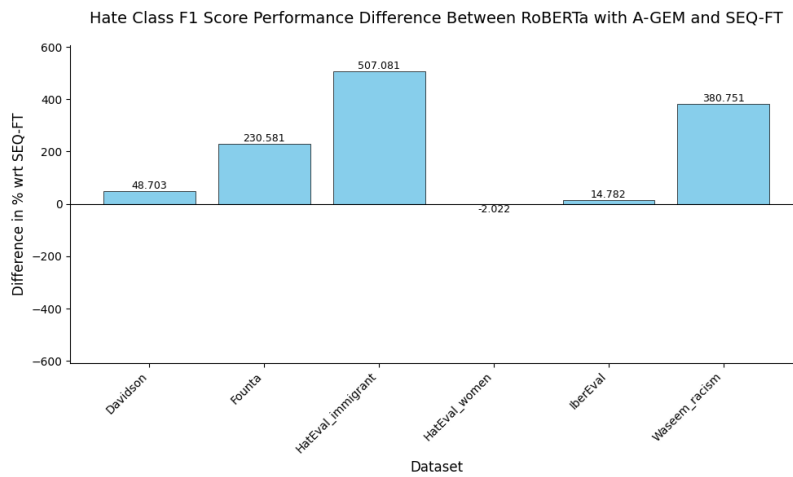
TABLE 3.26: Hate Class F1 Difference wrt. SEQ-FT (%)

Overall, A-GEM improved hate class predictions for every model, although the magnitude of this improvement is particularly striking for the longest experiment, Gen. $\rightarrow$ Alt. Mis. & Rac. In terms of models, RoBERTa and HateBERT are the models that experienced this effect to the highest degree.

Looking specifically at RoBERTa with A-GEM, the degree of improvement greatly depends on the datastream. Generally speaking, the longer the datastream, the greater cumulative difference there will be due to the degradation that SEQ-FT experiences. For the longest experiment, the performance of `HatEvalimmigrant` was more than five times better, `Waseemracism` was almost four times the performance of the SEQ-FT baseline and `Founta` was more than twice as better. The baseline struggles with `HatEvalimmigrant` and `Waseemracism` when they are positioned towards the end of the datastream. As we can see in the other experiments, the performance in the hate class of those datasets is much better when the datasets that come before are in their domain. Overall, RoBERTa A-GEM’s holds the hate-specific performance much better than the baseline in longer datasets.

Experiment Name	Testing Dataset	Hate F1 SEQ-FT	Hate F1 RoBERTa A-GEM
Gen. $\rightarrow$ Alt. Mis. & Rac.	Davidson	0.59	0.87
	Founta	0.16	0.54
	HatEval _{immigrant}	0.11	0.64
	HatEval _{women}	0.66	0.65
	IberEval	0.67	0.77
	Waseem _{racism}	0.09	0.44
Alt. Mis. & Rac.	Evalita	0.71	0.76
	HatEval _{immigrant}	0.62	0.62
	IberEval	0.91	0.94
	Waseem _{racism}	0.67	0.73
Rac. $\rightarrow$ Mis.	HatEval _{immigrant}	0.60	0.62
	IberEval	0.87	0.85
	Waseem _{racism}	0.64	0.73

TABLE 3.27: Hate Class F1 Performance

FIGURE 3.12: HATE Performance Difference w.r.t. SEQ-FT in Gen. $\rightarrow$ Alt. Mis. & Rac.

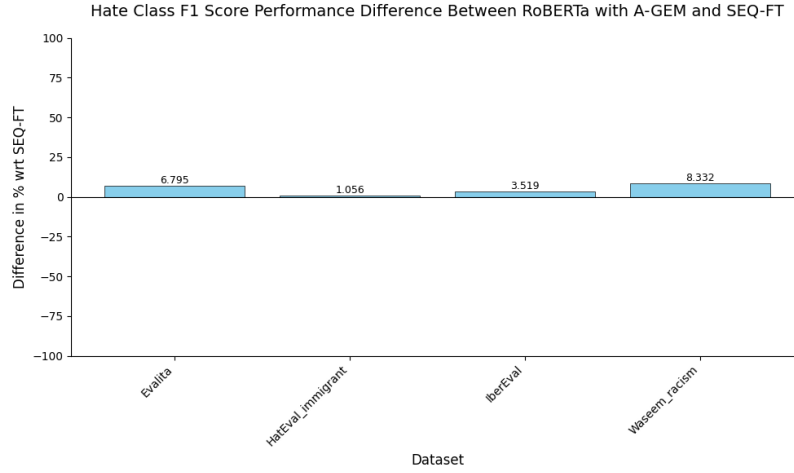


FIGURE 3.13: HATE Performance Difference w.r.t. SEQ-FT in Alt. Mis. & Rac.

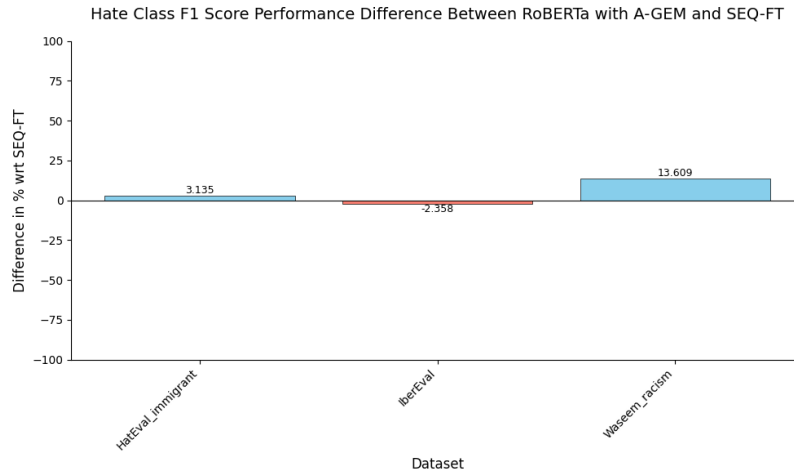


FIGURE 3.14: HATE Performance Difference w.r.t. SEQ-FT in Rac. → Mis.

3.8 Early Stopping and Continual Learning

The models were trained for eight epochs without early stopping with an AdamW optimizer. However, given the debate about whether early stopping is good practice or not, we tried training some continual learning models with that technique. Although most models did not need it, as their validation loss kept on decreasing, which indicates that they could have been trained for a bit longer, there were a couple of models that did stop early.

The general behavior observed was that if a model stopped early during any of the training stages, the validation from that point onward would hit a datastream global saddle point, that is, the validation in the following training stages would also get stuck and barely improve.

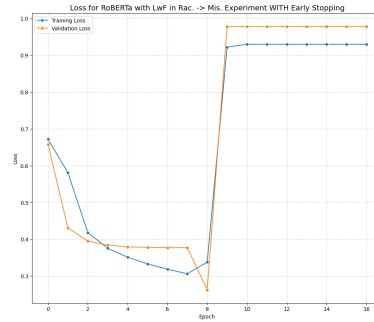


FIGURE 3.15: Example of the Early Stopping Loss Phenomena

Aside from the validation freeze, the fact that the model stopped at the lowest validation did not result in better performance. For example, RoBERTa with LwF in the Rac. $\rightarrow$ Mis. experiment, the LAST measure of the early stopper was 0.13 lower than its non-stopper counterpart. As to what could cause this phenomena, it may be the case that setting continual learning hyperparameters that favor staying closer to previous tasks parameters may produce an inevitable loss that is significantly higher than the part of the loss coming from the current task, effectively making the model stubborn against the new task. If this was the case, setting hyperparameters that favor plasticity rather than stability, for example lower EwC lambdas, could overcome these limitations and make early stopping feasible again. It could also be the case that continual learning is more forgiving to overfitting a model since part of that overfit will be forgotten if the continual learning setup is more prone to learning new tasks.

3.9 Considerations

It should be acknowledged that when using domain (in this case, Hate) specific models, one runs the risk of having models that have already seen the training data. However, the strongest of our models was not one pretrained on Hate Speech, but rather, the general RoBERTa.

One of the limitations of the continual learning techniques applied is that their effect should dilute as the streams become longer and longer. Although the implementation of A-GEM resets memory after every training stage, it is not uncommon to have continual learning setups where there is dedicated space (or parameters) for each past task instead of relying on the fact that the continuously regularized model will still meaningfully constrain the training in terms of the first tasks it learned. Moreover, the hyperparameters in the experiments were set following the techniques foundational papers when those may not be the optimal ones.

In terms of how many experiments one should run, an ideal scenario would be one where you can execute as many experiments as possible as well as their permutations so as to see how the positioning and pairs of datasets affect each other. However, it would be too computationally extensive, specially if one wants to consider several models and datastream lengths.

Chapter 4

Conclusion

4.1 Future Work

Continuing within the field of PLMs, we have seen that models can be sensitive to continual hyperparameters, as depending on how they are set, they may favor stability or plasticity. As far as we are aware, there has not been much research dealing with scheduling these parameters to adjust along the training datastreams.

After this overview of PLMs for Continual Learning, we have started working on methods for LLMs. Current methods have focused on ways to merge LoRA layers as an efficient way of adding new capabilities to LLMs but little work has focused on doing that in a continual learning way.

Another interesting direction to explore could be continual reinforcement learning for reasoning models by creating rationales from annotated implicit hate speech datasets such as ElSherief et al., 2021. A final direction we have embarked on is temporal continual learning and time-agnostic hate speech detection ¹.

4.2 Conclusion

All in all, we have explored four continual learning techniques for Hate Speech Detection with PLMs three different PLMs in three different datastreams. We have shown how sequential trainings may outperform merged datasets if done in a meta-learning kind of way (whether intentionally or unintentionally) and how the data replay and gradient projection method, A-GEM, is able to outperform vanilla sequential training, and the other continual learning techniques, in long sequences as well as improving the prediction of hateful messages. Previous training on hate-specific corpus did not necessarily give advantage to fBERT and HateBERT as RoBERTa A-GEM consistently was the best performer for each experiment despite its more general nature.

¹For more information, refer to *Introduction to Introductory Survey on Temporal Hate Speech*.

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Appendix A

Continual Learning Overview

Continual Learning is subfield of Deep Learning whose main objective is to improve a model’s ability to learn sequentially, that is, in data streams, rather than in a stationary, one-and-done fashion (Hadsell et al., 2020). The main motivation for Continual Learning is training models that can rapidly adapt, either to new tasks, new data distributions or new classes without forgetting the knowledge they had previously acquired during earlier stages of their training.

There are several challenges and limitations associated with Continual Learning. The most basic limitation is that the training data of a previous stage is assumed not to be accessible after said training stage, making retraining a model with all the accumulated data effectively impossible (Ke & Liu, 2023). It follows from this limitation that in a pragmatic continual learning scenario the order of the data streams cannot be predicted since the future data would not be accessible at the present time of starting the training.

The main challenge that Continual Learning systems face is the *stability-plasticity* dilemma (Parisi et al., 2019). As models acquire new information, they should avoid *catastrophic forgetting*, the sudden and steep loss in performance of previous tasks when training on a new task or data stream, and they should be able to incorporate new information rapidly and effectively. That is, their performance in previous tasks should remain stable but it should also be elastic enough to quickly adapt to new tasks (Wang et al., 2024).

A.1 Continual Learning Approaches

Wang et al., 2024 identifies several major approaches to Continual Learning:

1. **Regularization-Based Approaches** involve using regularization terms to constrain how far the parameters move with respect to the model learned in the previous task. The term usually depends on the importance associated with each parameter, which can be calculated by importance measures, like the Fisher Information Matrix (FIM) in the case of Elastic Weight Regularization (EWC) (Kirkpatrick et al., 2017) or accounting for the the sensitivity of predictive results to parameter changes in the case of Memory Aware Synapses (MAS) (Aljundi et al., 2018).
2. **Replay-Based Approaches** consist of replaying a small amount of samples from previous experiences stored in a memory buffer or training a generative model, like a Generative Adversarial Networks (GAN) (Goodfellow et al., 2014) or Variational Auto-Encoders (VAE) (Kingma & Welling, 2022), to replay generated data without the need to store samples. Averaged GEM (A-GEM) (Chaudhry et al., 2019), an improved version of Gradient Episodic Memory

(GEM) (Lopez-Paz & Ranzato, 2022), aims to ensure that at every training step the episodic memory loss over the previous tasks does not increase by computing the gradient of a random subset of memory examples and projecting gradient updates towards those reference gradients.

3. Optimization-Based Approaches aim to perform gradient projection to constrain parameter updates to the orthogonal direction of previous input spaces.
4. Architecture-Based Approaches construct task-specific parameters for different tasks rather than having tasks share the same parameters.

A.1.1 Continual Learning Techniques Overview

A.1.1.1 EWC

Elastic weight consolidation (EWC) (Kirkpatrick et al., 2017) is a continual learning regularization technique that aims to constrain parameter updates that were important to a previous task. EWC anchors the parameters update of the new task to the space of the old, already learned task.

In EWC, the parameter importance is represented by the diagonal of its Fisher Information Matrix and the penalty is applied per parameter, unlike other methods that use a global penalization.

$$\mathcal{L}(\theta) = \mathcal{L}_B(\theta) + \sum_i \frac{\lambda}{2} F_i (\theta_i - \theta_{A,i}^*)^2$$

FIGURE A.1: EWC Loss Function (Kirkpatrick et al., 2017)

A.1.1.2 MAS

Memory Aware Synapses (MAS) (Aljundi et al., 2018) is another continual learning regularization technique. It also functions by applying an individual penalty to each parameter based on its importance.

Parameter importance in MAS is represented by the sensitivity of the output to the parameter's change. This is the main difference between EWC and MAS: MAS is an unsupervised method while EWC is supervised as it requires the Fisher Information Matrix, which represents how much a parameter change modifies the loss, to be computed.

$$L(\theta) = L_n(\theta) + \lambda \sum_{i,j} \Omega_{ij} (\theta_{ij} - \theta_{ij}^*)^2$$

FIGURE A.2: MAS Loss Function (Aljundi et al., 2018)

A.1.1.3 A-GEM

Averaged Gradient Episodic Memory (A-GEM) (Chaudhry et al., 2019) is a replay and optimization continual learning technique. Broadly, it works by initiating a memory buffer to store data from previous tasks, replays it to compute gradients for the old task and uses them as constrain such that, if the gradients for the current data and the reference gradients have a negative dot product, the old gradients are projected by A.3.

$$\tilde{g} = g - \frac{g^\top g_{ref}}{g_{ref}^\top g_{ref}} g_{ref}$$

FIGURE A.3: A-GEM Gradient Projection (Chaudhry et al., 2019)

```

1: procedure TRAIN( $f_\theta, \mathcal{D}^{train}, \mathcal{D}^{test}$ )
2:    $\mathcal{M} \leftarrow \{\}$ 
3:    $A \leftarrow 0 \in \mathbb{R}^{T \times T}$ 
4:   for  $t = \{1, \dots, T\}$  do
5:     for  $(\mathbf{x}, y) \in \mathcal{D}_t^{train}$  do
6:        $(\mathbf{x}_{ref}, y_{ref}) \sim \mathcal{M}$ 
7:        $g_{ref} \leftarrow \nabla_{\theta} \ell(f_\theta(\mathbf{x}_{ref}, t), y_{ref})$ 
8:        $g \leftarrow \nabla_{\theta} \ell(f_\theta(\mathbf{x}, t), y)$ 
9:       if  $g^\top g_{ref} \geq 0$  then
10:         $\tilde{g} \leftarrow g$ 
11:       else
12:         $\tilde{g} \leftarrow g - \frac{g^\top g_{ref}}{g_{ref}^\top g_{ref}} g_{ref}$ 
13:       end if
14:        $\theta \leftarrow \theta - \alpha \tilde{g}$ 
15:     end for
16:      $\mathcal{M} \leftarrow \text{UPDATEEPSMEM}(\mathcal{M}, \mathcal{D}_t^{train}, T)$ 
17:      $A_{t,:} \leftarrow \text{EVAL}(f_\theta, \mathcal{D}^{test})$ 
18:   end for
19:   return  $f_\theta, A$ 
20: end procedure

```

FIGURE A.4: A-GEM Training Algorithm (Chaudhry et al., 2019)

A.1.1.4 LwF

Finally, Learning without Forgetting (LwF) (Li & Hoiem, 2017) is a knowledge distillation continual learning technique. Simply put, LwF saves a copy of the model after the task has finished training to use as the teacher and it is used to compute a distillation loss based on the current task batch by calculating the KL divergence between the pseudo-probabilities of teacher (old model) and the student (model during current task). Finally, it applies a temperature hyperparameter to the softmax and a lambda to balance the loss at current time and the distillation loss.

A.2 Continual Learning Scenarios

Depending on whether the tasks are clearly delimited during the training and whether the data for a task is available before training or arriving in batches, there are several Continual Learning scenarios (Wang et al., 2024).

Yang et al., 2024 classifies the offline and online settings in the following manner: at a first level, the scenarios can be distinguished between those set ups where each task is delimited with its own dataset which is already available before training (Offline CL) and those where the dataset for each task is not fully available before starting training, that is, models are trained with a continuous data flow of singular examples or mini-batches (Online CL)

- Offline Learning
 - Task-Incremental Learning: each task has its own unique objective and the classes in the tasks are not necessarily the same. The task ids are available during both the training and the testing of the model.

- Domain-Incremental Learning: the objective of the tasks is the same but the data for each task has a different distribution. Task ids are not necessary and the classes of the tasks are the same.
- Class-Incremental Learning: the tasks can have the same objective but the classes for each tasks are different and the task ids are available during training.
- Online Learning
 - Hard Task Boundary: the arrival of tasks is structured; there is no overlap of data from different tasks arriving simultaneously.
 - Blurry Task Boundary: the tasks are not clearly differentiated and the data for each task is intermixed with other data from different tasks.

Additionally, Wang et al., 2024 mentions Instance-Incremental Learning, the task is the same and the data arrives in batches and Continual Pre-training, where the pre-training data arrives sequentially and the main goal of the training is to improve knowledge transfer to downstream tasks.

A.3 Continual Learning Metrics

The evaluation protocol for Continual Learning models is slightly different from other learning paradigms. Because learning plasticity and stability are crucial, not only are models tested in the current dataset they are being trained on, but also in every dataset part of the data stream, after each training stage, as well as the model trained and tested on each task independently (Yang et al., 2024). This way we obtain a matrix where the columns represent the test task and the rows represent the training task; each column represents the evolution of the performance of the model on a particular dataset across the training data stream and each row represents the performance of the model on every dataset after training up to that point in the data stream. The diagonal represents the performance of the model on a dataset after being trained up to the point where that dataset appears in the data stream.

There are three main types of metrics used to evaluate Continual Learning models:

1. Overall performance: assesses the performance across all tasks.
2. Memory stability: focuses on the ability to retain knowledge.
3. Learning plasticity: evaluates the ability of the model to learn new tasks.

Overall Performance Metrics

- Last is the average performance of all the tasks after finishing training.

$$Last = \frac{1}{N} \sum_{i=1}^N R_{N,i}$$

- Average Incremental Accuracy (AIA) measures the progression of the tasks across time after training the model on them. The tasks that have not been yet seen in training are ignored.

$$AIA = \frac{1}{N} \sum_{i=1}^N \left(\frac{1}{i} \sum_{j=1}^i R_{i,j} \right)$$

- Transfer addresses how the previous tasks influence future tasks.

$$Transfer = \frac{1}{N-1} \sum_{i=2}^N \left(\frac{1}{i-1} \sum_{j=1}^{i-1} R_{j,i} \right)$$

Memory stability Metrics

- Backward Transfer (BWT) measures how much a model forgets about an old task after training on a new task.

$$BWT = \frac{1}{N-1} \sum_{i=1}^{N-1} R_{N,i} - R_{i,i}$$

Learning plasticity Metrics

- Forward Transfer (FWT) quantifies how much the performance of a future task improves by training on a previous task before reaching said task.

$$FWT = \frac{1}{N-1} \sum_{i=2}^N R_{i-1,i} - R_{0,i}$$

Appendix B

Hate Speech Datasets Contamination

B.1 Domain Transfer Experiments Datasets Contamination

B.1.1 Contamination of Experiment Data streams by Individual Datasets

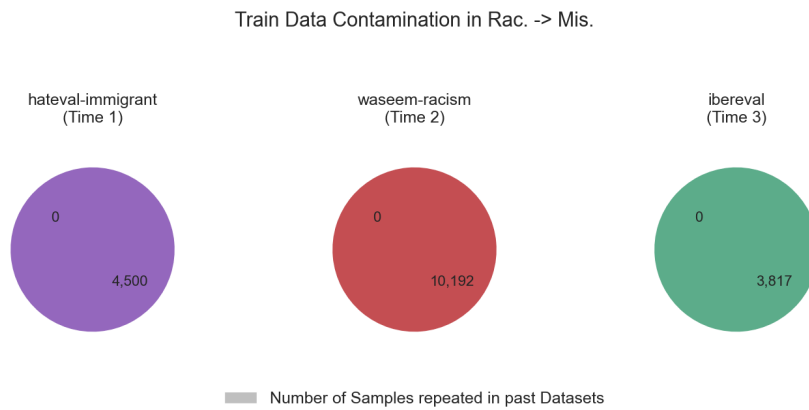


FIGURE B.1: Train Data Repetition in Rac. → Mis.

B.1.1.1 Experiment Alt. Mis. & Rac.

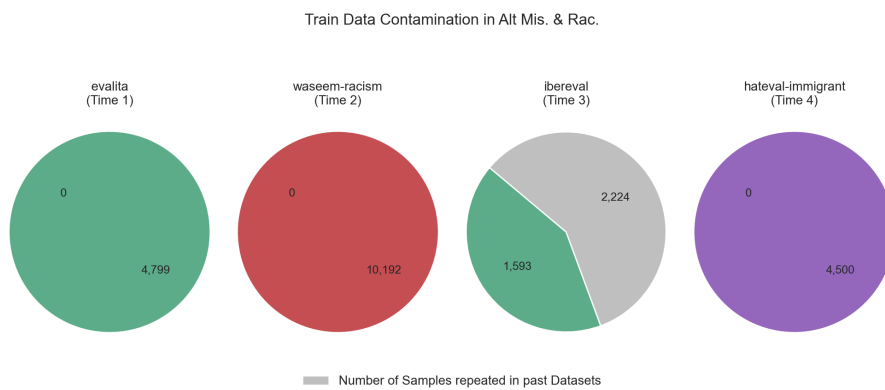


FIGURE B.2: Train Data Repetition in Alt. Mis. & Rac.

B.1.1.2 Experiment Gen. $\rightarrow$ Alt. Mis. & Rac.

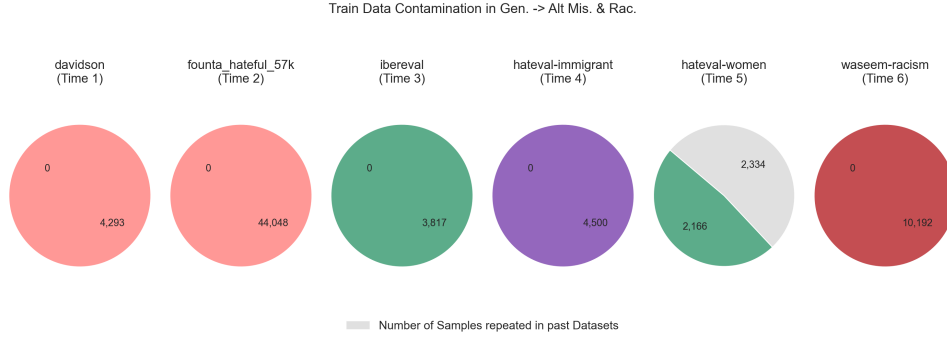


FIGURE B.3: Train Data Repetition in Gen. $\rightarrow$ Alt. Mis. & Rac.

B.1.2 Cumulative Contamination by Experiment Data stream

B.1.2.1 Experiment Rac. $\rightarrow$ Mis.

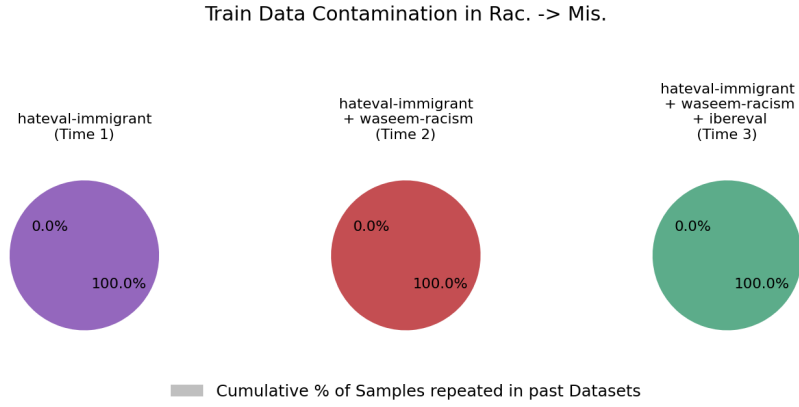


FIGURE B.4: Cumulative Train Data Repetition in Rac. $\rightarrow$ Mis.

B.1.2.2 Experiment Alt. Mis. & Rac.

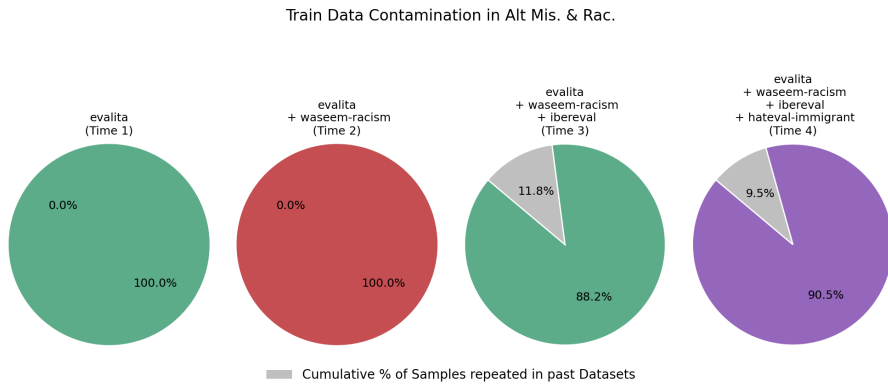
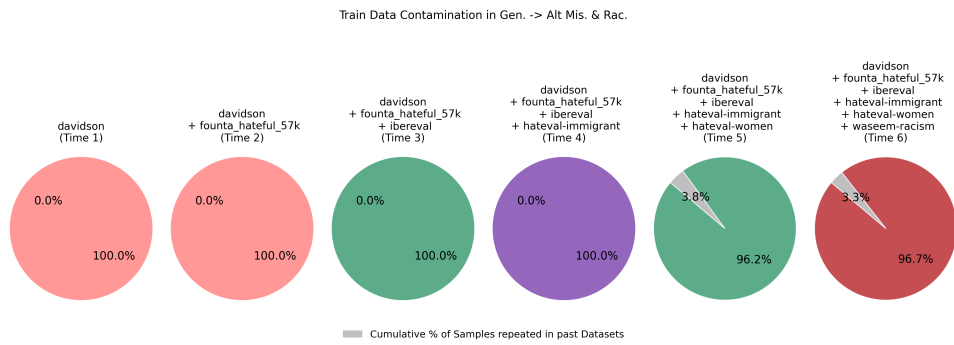


FIGURE B.5: Cumulative Train Data Repetition in Alt. Mis. & Rac.

B.1.2.3 Experiment Gen. $\rightarrow$ Alt. Mis. & Rac.FIGURE B.6: Cumulative Train Data Repetition in Gen. $\rightarrow$ Alt. Mis. & Rac.

Appendix C

Results for the Hyperparameter Search

Aside from the previously discussed A-GEM memory size proportion experiments, we also performed hyperparameter experiments for other models and continual learning techniques and report the results.

Experiment	Model	CL Technique	Hyperparams	CL Metrics				
				LAST	Average Incremental F1	Transfer	BWT	FWT
Gen. $\rightarrow$ Alt. Mis. & Rac.	RoBERTa	A-GEM	% of train data in Memory: 1%	0.72	0.80	0.81	-0.03	0.38
			% of train data in Memory: 2.5%	0.72	0.81	0.82	-0.02	0.38
			SEQ-FT	0.61	0.75	0.74	-0.12	0.33
	HateBERT	MAS	MAS Lambda: 1000	0.66	0.74	0.76	-0.00	0.18
			MAS Lambda: 1500	0.64	0.72	0.74	0.01	0.16
			MAS Lambda: 500	0.64	0.74	0.75	-0.02	0.15
	fBERT	EWC	SEQ-FT	0.63	0.72	0.71	-0.07	0.09
			EWC Lambda: 1000	0.65	0.76	0.77	-0.04	-0.02
			EWC Lambda: 1500	0.65	0.76	0.77	-0.04	-0.02
			EWC Lambda: 2000	0.65	0.76	0.78	-0.02	-0.03
			SEQ-FT	0.69	0.77	0.78	-0.01	-0.02
Alt. Mis. & Rac.	RoBERTa	A-GEM	% of train data in Memory: 1%	0.74	0.81	0.81	-0.05	0.29
			% of train data in Memory: 2.5%	0.76	0.82	0.84	-0.02	0.30
			% of train data in Memory: 5%	0.76	0.83	0.85	-0.02	0.30
			SEQ-FT	0.70	0.75	0.67	-0.09	0.19
	fBERT	A-GEM	% of train data in Memory: 1%	0.71	0.74	0.75	0.00	-0.09
			% of train data in Memory: 2.5%	0.72	0.75	0.76	0.00	-0.07
			% of train data in Memory: 5%	0.72	0.74	0.75	0.02	-0.06
			SEQ-FT	0.68	0.69	0.64	-0.02	-0.15
Rac. $\rightarrow$ Mis.	RoBERTa	A-GEM	% of train data in Memory: 1%	0.74	0.60	0.57	0.04	0.31
			% of train data in Memory: 2.5%	0.75	0.60	0.56	0.06	0.31
			% of train data in Memory: 5%	0.75	0.61	0.56	0.02	0.27
			SEQ-FT	0.75	0.63	0.64	0.03	0.28
	fBERT	A-GEM	% of train data in Memory: 1%	0.73	0.62	0.61	0.04	-0.16
			% of train data in Memory: 2.5%	0.73	0.63	0.61	0.03	-0.17
			% of train data in Memory: 5%	0.73	0.63	0.61	0.03	-0.17
			SEQ-FT	0.71	0.62	0.60	0.02	-0.17

TABLE C.1: Continual Learning Hyperparam Experiment Results

Appendix D

LLM Prompting

D.1 Experimental Setup

Testing Method	Sampling	N. Shots
Zero-shot Prompting	N/A	0
Few-shot Prompting	Random Sampling	2
		6
		10

TABLE D.1: LLM Baselines

D.2 Prompts

D.2.1 Zero Shot Prompt

You are a social media content moderator.

INSTRUCTION: The following is a social media message that needs to be classified with the label HATEFUL or NOT HATEFUL.

MESSAGE:

OUTPUT AND FORMAT: your output should be just the HATEFUL or NOT HATEFUL label.

LABEL:

D.2.2 Few Shot Prompt

You are a social media content moderator.

INSTRUCTION: The following is a set of social media messages examples already classified with the label HATEFUL or NOT HATEFUL followed by the social media message you have to classify with the label HATEFUL or NOT HATEFUL.

Example MESSAGE:

Example LABEL:

YOUR MESSAGE:

OUTPUT AND FORMAT: your output should be just the HATEFUL or NOT HATEFUL label classification for YOUR MESSAGE.

LABEL:

D.3 Results

Model	Testing Dataset	F1 Score
Llama-3.2-1B-Instruct	Evalita	0.23
	IberEval	0.25
	HatEval _{immigrant}	0.24
	Davidson	0.28
	Founta	0.31
	HatEval _{women}	0.24
	Waseem _{racism}	0.29
	Waseem _{sexism}	0.29
	Evalita	0.23
	IberEval	0.26
	HatEval _{immigrant}	0.22
	Davidson	0.24
	Founta	0.19
	HatEval _{women}	0.21
	Waseem _{racism}	0.18
	Waseem _{sexism}	0.20
Llama-Guard-3-1B	Evalita	0.19
	IberEval	0.31
	HatEval _{immigrant}	0.21
	Davidson	0.43
	Founta	0.33
	HatEval _{women}	0.22
	Waseem _{racism}	0.44
	Waseem _{sexism}	0.28
Qwen3-1.7B	Davidson	0.56
	Evalita	0.49
	Founta	0.47
	HatEval _{immigrant}	0.57
	HatEval _{women}	0.49
	IberEval	0.53
	Waseem _{racism}	0.50
	Waseem _{sexism}	0.47
Qwen3-4B-Instruct-2507	Evalita	0.53
	IberEval	0.67
	HatEval _{immigrant}	0.51
	Davidson	0.66
	Founta	0.61
	HatEval _{women}	0.54
	Waseem _{racism}	0.58
	Waseem _{sexism}	0.50

TABLE D.2: Zero Shot Prompting

Model	Testing Dataset	N. Shots	F1 Score
Llama-3.2-1B-Instruct	Evalita	10	0.41

Llama-3.2-3B-Instruct	IberEval	10	0.57
	HatEval _{immigrant}	10	0.42
	Davidson	10	0.68
	Founta	10	0.42
	HatEval _{women}	10	0.26
	Waseem _{racism}	10	0.31
	Waseem _{sexism}	10	0.50
	Evalita	2	0.49
	IberEval	2	0.46
	HatEval _{immigrant}	2	0.50
	Davidson	2	0.33
	Founta	2	0.39
	HatEval _{women}	2	0.41
	Waseem _{racism}	2	0.37
	Waseem _{sexism}	2	0.44
	Evalita	6	0.39
	IberEval	6	0.41
	HatEval _{immigrant}	6	0.46
	Davidson	6	0.64
	Founta	6	0.55
	HatEval _{women}	6	0.52
	Waseem _{racism}	6	0.52
	Waseem _{sexism}	6	0.49
	Evalita	10	0.09
Qwen3-1.7B	IberEval	10	0.22
	HatEval _{immigrant}	10	0.30
	Davidson	10	0.19
	Founta	10	0.13
	HatEval _{women}	10	0.20
	Waseem _{racism}	10	0.27
	Waseem _{sexism}	10	0.16
	Evalita	6	0.22
	IberEval	6	0.25
	HatEval _{immigrant}	6	0.20
	Davidson	6	0.20
	Founta	6	0.07
	HatEval _{women}	6	0.20
	Waseem _{racism}	6	0.13
	Waseem _{sexism}	6	0.14
	Evalita	2	0.22
	IberEval	2	0.22
	HatEval _{immigrant}	2	0.20
	Davidson	2	0.12
	Founta	2	0.07
	HatEval _{women}	2	0.07
	Waseem _{racism}	2	0.15
	Waseem _{sexism}	2	0.12
	Evalita	10	0.35
	IberEval	10	0.38
	HatEval _{immigrant}	10	0.25
	Davidson	10	0.43

	Founta	10	0.32
	HatEval _{women}	10	0.37
	Waseem _{racism}	10	0.46
	Waseem _{sexism}	10	0.29
	Evalita	2	0.23
	IberEval	2	0.27
	HatEval _{immigrant}	2	0.38
	Davidson	2	0.31
	Founta	2	0.33
	HatEval _{women}	2	0.37
	Waseem _{racism}	2	0.31
	Waseem _{sexism}	2	0.44
	Evalita	6	0.25
	IberEval	6	0.41
	HatEval _{immigrant}	6	0.41
	Davidson	6	0.44
	Founta	6	0.48
	HatEval _{women}	6	0.37
	Waseem _{racism}	6	0.46
	Waseem _{sexism}	6	0.45
Qwen3-4B-Instruct-2507	Evalita	2	0.36
	IberEval	2	0.47
	HatEval _{immigrant}	2	0.36
	Davidson	2	0.51
	Founta	2	0.33
	HatEval _{women}	2	0.33
	Waseem _{racism}	2	0.35
	Waseem _{sexism}	2	0.27
	Evalita	10	0.30
	IberEval	10	0.46
	HatEval _{immigrant}	10	0.23
	Davidson	10	0.39
	Founta	10	0.39
	HatEval _{women}	10	0.27
	Waseem _{racism}	10	0.39
	Waseem _{sexism}	10	0.33
	Evalita	6	0.30
	IberEval	6	0.50
	HatEval _{immigrant}	6	0.29
	Davidson	6	0.42
	Founta	6	0.37
	HatEval _{women}	6	0.27
	Waseem _{racism}	6	0.35
	Waseem _{sexism}	6	0.20

TABLE D.3: Few-shot Prompting

Appendix E

Prediction Sequences

E.1 Experiment Rac. $\rightarrow$ Mis.

Time 1 HatEval_{immigrant}	Time 2 Waseem_{racism}	Time 3 IberEval	Proportion
Incorrect	Incorrect	Incorrect	0.10
		Correct	0.07
	Correct	Incorrect	0.01
		Correct	0.08
Correct	Correct	Correct	0.56
		Incorrect	0.05
	Incorrect	Correct	0.11
		Incorrect	0.01

TABLE E.1: Rac. $\rightarrow$ Mis. Prediction Sequence Proportion for Hateful Messages

Time 1 HatEval_{immigrant}	Time 2 Waseem_{racism}	Time 3 IberEval	Proportion
Incorrect	Incorrect	Incorrect	0.18
		Correct	0.05
	Correct	Incorrect	0.00
		Correct	0.06
Correct	Correct	Correct	0.68
		Incorrect	0.02
	Incorrect	Correct	0.01
		Incorrect	0.01

TABLE E.2: Rac. $\rightarrow$ Mis. Prediction Sequence Proportion for Not Hateful Messages

E.2 Experiment Alt. Mis. & Rac.

Time 1 Evalita	Time 2 Waseem_{racism}	Time 3 IberEval	Time 4 HatEval_{immigrant}	Proportion
Incorrect	Incorrect	Incorrect	Incorrect	0.08
			Correct	0.29
		Correct	Incorrect	0.00
			Correct	0.01
	Correct	Incorrect	Incorrect	0.00
			Correct	0.03
		Correct	Incorrect	0.01
			Correct	0.21
Correct	Correct	Correct	Correct	0.32
			Incorrect	0.00
		Incorrect	Correct	0.01
			Incorrect	0.00
	Incorrect	Correct	Correct	0.02
			Incorrect	0.00
		Incorrect	Correct	0.01
			Incorrect	0.01

TABLE E.3: Alt. Mis. & Rac. Prediction Sequence Proportion for Hateful Messages

Time 1 Evalita	Time 2 Waseem_{racism}	Time 3 IberEval	Time 4 HatEval_{immigrant}	Proportion
Incorrect	Incorrect	Incorrect	Incorrect	0.02
			Correct	0.00
		Correct	Incorrect	0.00
			Correct	0.00
	Correct	Incorrect	Incorrect	0.00
			Correct	0.00
		Correct	Incorrect	0.00
			Correct	0.02
Correct	Correct	Correct	Correct	0.72
			Incorrect	0.18
		Incorrect	Correct	0.00
			Incorrect	0.01
	Incorrect	Correct	Correct	0.00
			Incorrect	0.01
		Incorrect	Correct	0.00
			Incorrect	0.03

TABLE E.4: Alt. Mis. & Rac. Prediction Sequence Proportion for Not Hateful Messages

E.3 Experiment Gen. $\rightarrow$ Alt. Mis. & Rac.

Time 1	Time 2	Time 3	Time 4	Time 5	Time 6	Proportion
Davidson	Founta	IberEval	HatEval _{immigrant}	HatEval _{women}	Waseem _{racism}	
Incorrect	Incorrect	Incorrect	Incorrect	Incorrect	Incorrect	0.22
				Correct	Correct	0.02
			Correct	Incorrect	Incorrect	0.03
				Incorrect	Correct	0.04
				Correct	Correct	0.02
				Correct	Correct	0.06
		Correct	Correct	Incorrect	Correct	0.00
				Correct	Correct	0.00
	Correct	Incorrect	Incorrect	Incorrect	Incorrect	0.00
			Correct	Incorrect	Incorrect	0.02
				Correct	Correct	0.02
				Correct	Correct	0.04
		Correct	Correct	Incorrect	Incorrect	0.01
				Correct	Correct	0.01
				Correct	Correct	0.03
Correct	Correct	Correct	Correct	Correct	Correct	0.39
				Incorrect	Correct	0.04
					Incorrect	0.02
		Incorrect	Correct	Correct	Correct	0.00
				Incorrect	Correct	0.00
					Incorrect	0.01
	Incorrect	Correct	Correct	Correct	Correct	0.00
				Incorrect	Correct	0.00
		Incorrect	Correct	Correct	Correct	0.00
				Incorrect	Correct	0.00
			Incorrect	Incorrect	Incorrect	0.00
				Incorrect	Incorrect	0.00

TABLE E.5: Gen. $\rightarrow$ Alt. Mis. & Rac. Prediction Sequence Proportion for Hateful Messages

Time 1	Time 2	Time 3	Time 4	Time 5	Time 6	Proportion
Davidson	Founta	IberEval	HatEval _{immigrant}	HatEval _{women}	Waseem _{racism}	
Incorrect	Incorrect	Incorrect	Incorrect	Incorrect	Incorrect	0.03
				Correct	Incorrect	0.01
					Correct	0.01
		Correct	Incorrect	Incorrect	Incorrect	0.00
				Correct	Incorrect	0.00
					Correct	0.00
	Correct	Incorrect	Incorrect	Incorrect	Incorrect	0.00
				Correct	Incorrect	0.00
					Correct	0.00
		Correct	Incorrect	Incorrect	Incorrect	0.00
				Correct	Incorrect	0.00
					Correct	0.00
			Correct	Correct	Correct	0.00
Correct	Correct	Correct	Correct	Correct	Correct	0.86
					Incorrect	0.01
				Incorrect	Incorrect	0.01
			Incorrect	Correct	Correct	0.03
					Incorrect	0.00
				Incorrect	Incorrect	0.01
		Incorrect	Incorrect	Correct	Correct	0.00
					Incorrect	0.00
	Incorrect	Correct	Correct	Incorrect	Incorrect	0.00
			Incorrect	Correct	Correct	0.00
				Correct	Correct	0.01
					Incorrect	0.00
				Incorrect	Incorrect	0.00
		Incorrect	Incorrect	Correct	Correct	0.00
				Incorrect	Incorrect	0.00

TABLE E.6: Gen. $\rightarrow$ Alt. Mis. & Rac. Prediction Sequence Proportion for Not Hateful Messages

Appendix F

Clustering Implementation

The algorithm used to cluster the prediction sequences was K-Medoids. Given that our sequences are binary, either a prediction is Correct or Incorrect, we use Hamming Distance as our distance measure and we weight the start and end of the sequences separately, with alpha and beta hyperparameters, to differentiate between sequences whose start is the same but the ending is different from sequences whose start is different but the ending is the same. We set the beginning and end of a sequence manually for each experiment. As far as the number of clusters, we set a range of 3 to 6 clusters to try and select the final number of clusters by computing the within-cluster weighted Hamming Distance for each cluster number and finding the Elbow point. Finally, we take the medoids to represent the most representative prediction sequences for each experiment.

Experiment	Start	Ending	Experiment Length	Alpha Beginning	Beta Ending	Number of Clusters
Gen. → Alt. Mis. & Rac.	1-2	5-6	6	1.25	1.5	6
Alt. Mis. & Rac.	1	4	4	1.25	1.5	4
Rac. → Mis.	1	3	3	1.25	1.5	4

TABLE F.1: Clustering Implementation Details